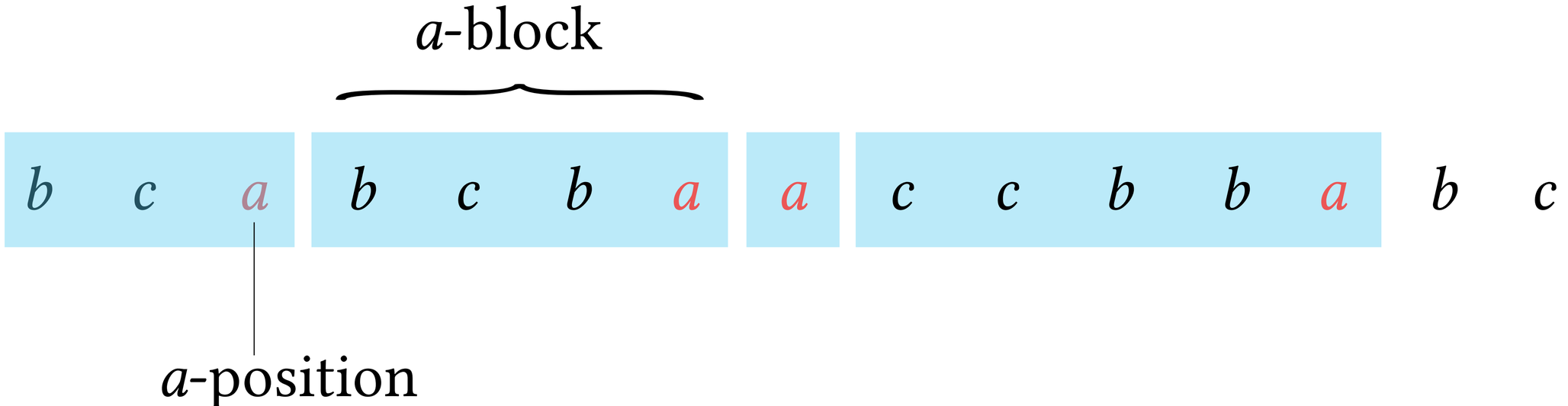
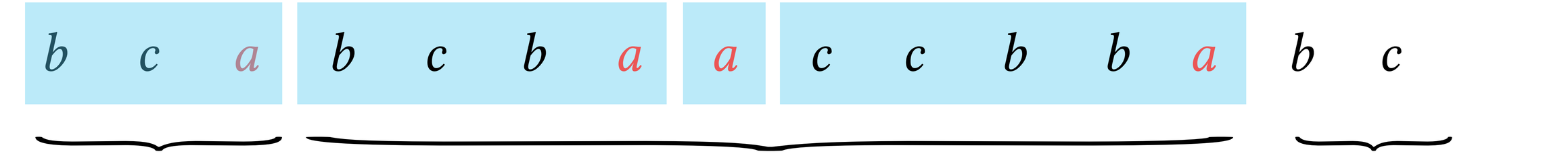
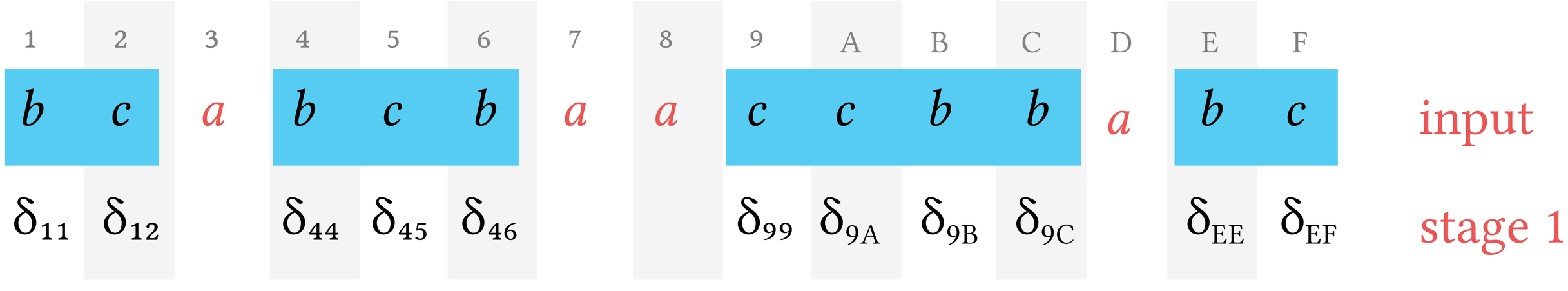
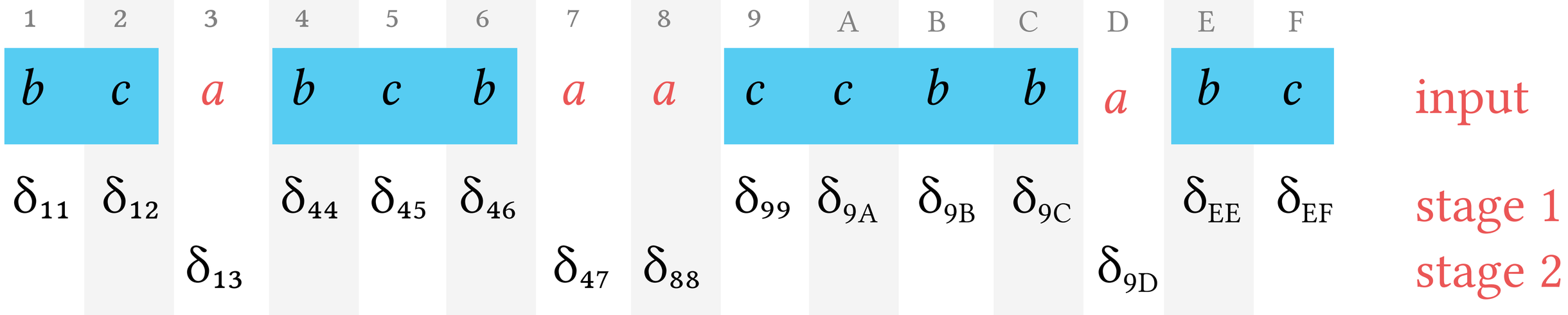
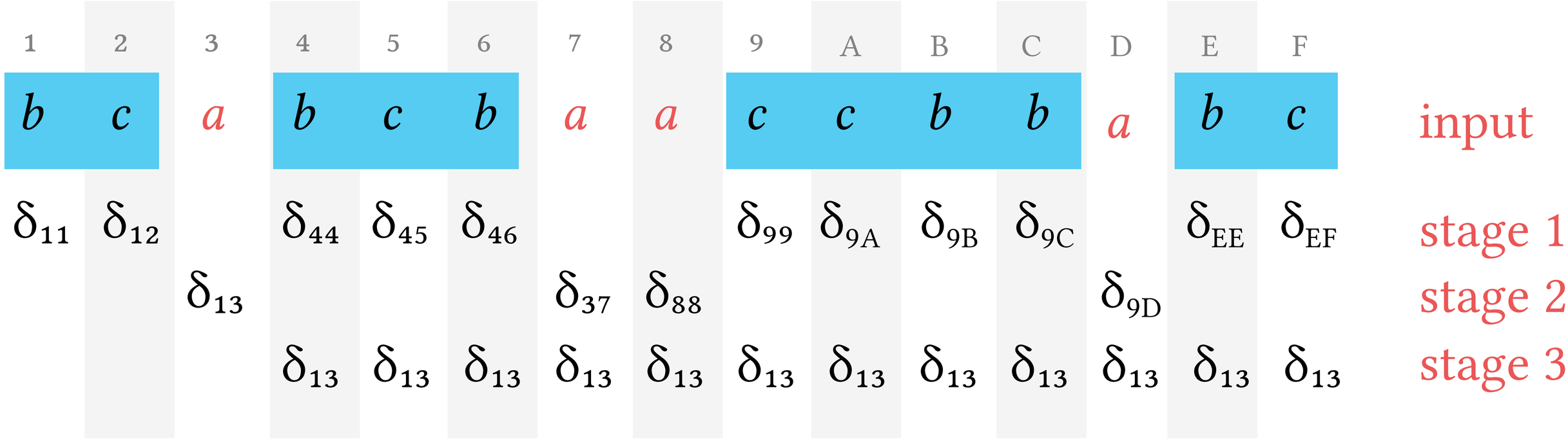
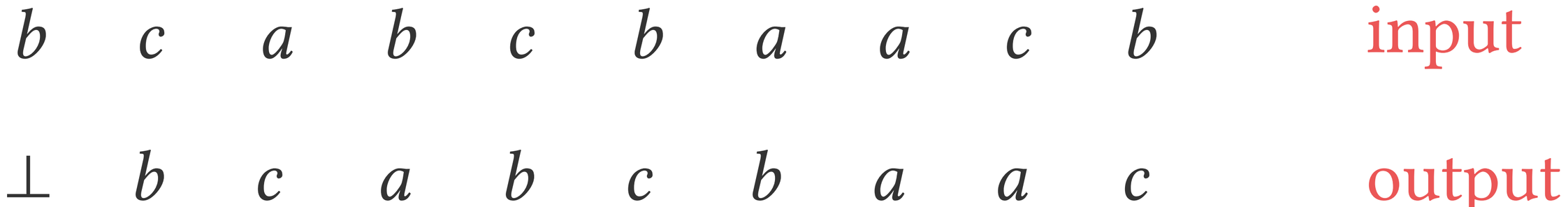

a-block
b c a b c b a a c c b b a b c
a-position

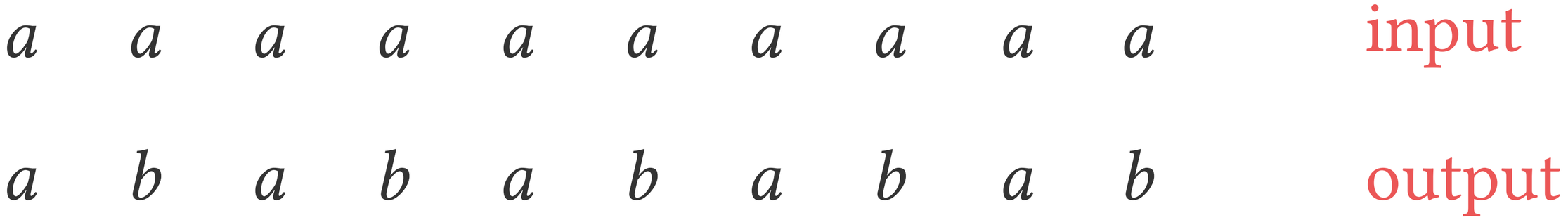

b c a
b c b a
a
c c b b a
b c
first part:
first a-block
middle part:
all a-blocks except the first
last part:
a-free suffix

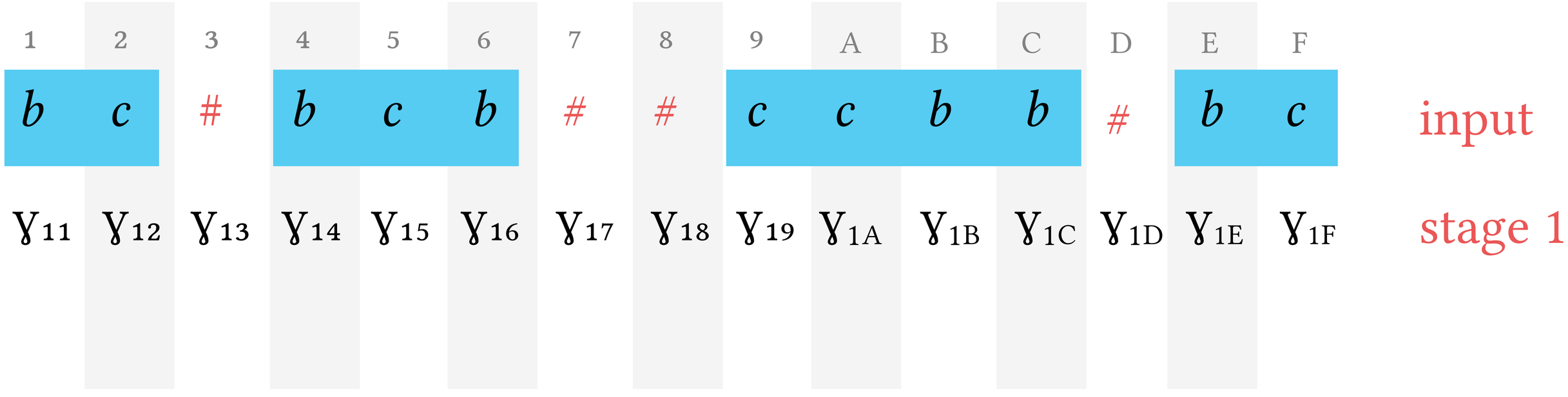

1 2 3 4 5 6 7 8 9 A B C D E F
b c a b c b a a c c b b a b c
input
δ11 δ12 δ44 δ45 δ46 δ99 δ9A δ9B δ9C δEE δEF
stage 1

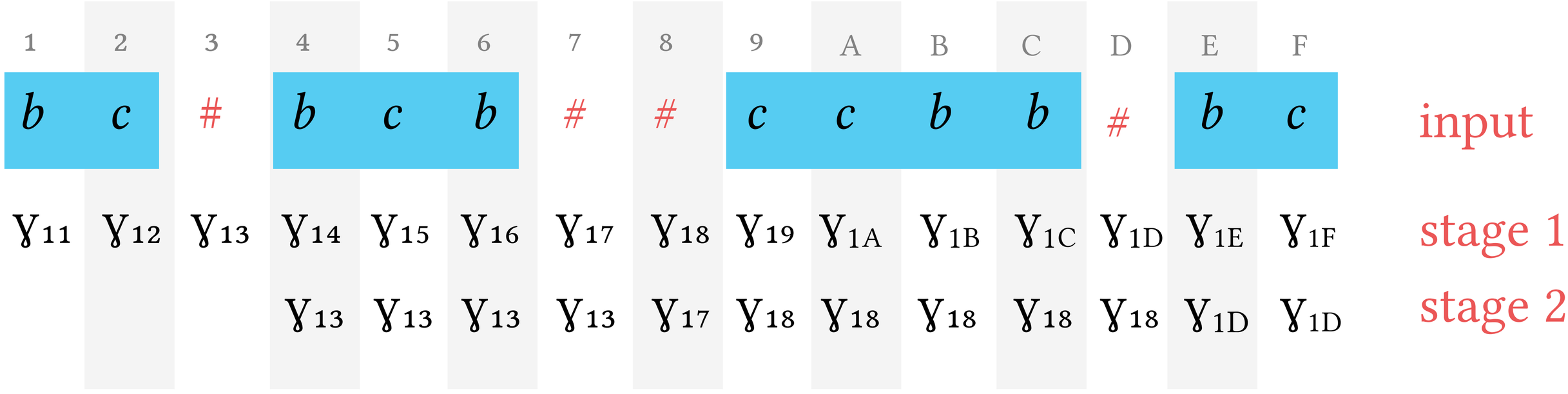

1 2 3 4 5 6 7 8 9 A B C D E F
b c a b c b a a c c b b a b c
input
δ11 δ12 δ44 δ45 δ46 δ99 δ9A δ9B δ9C δEE δEF
stage 1
δ13 δ47 δ88 δ9D
stage 2

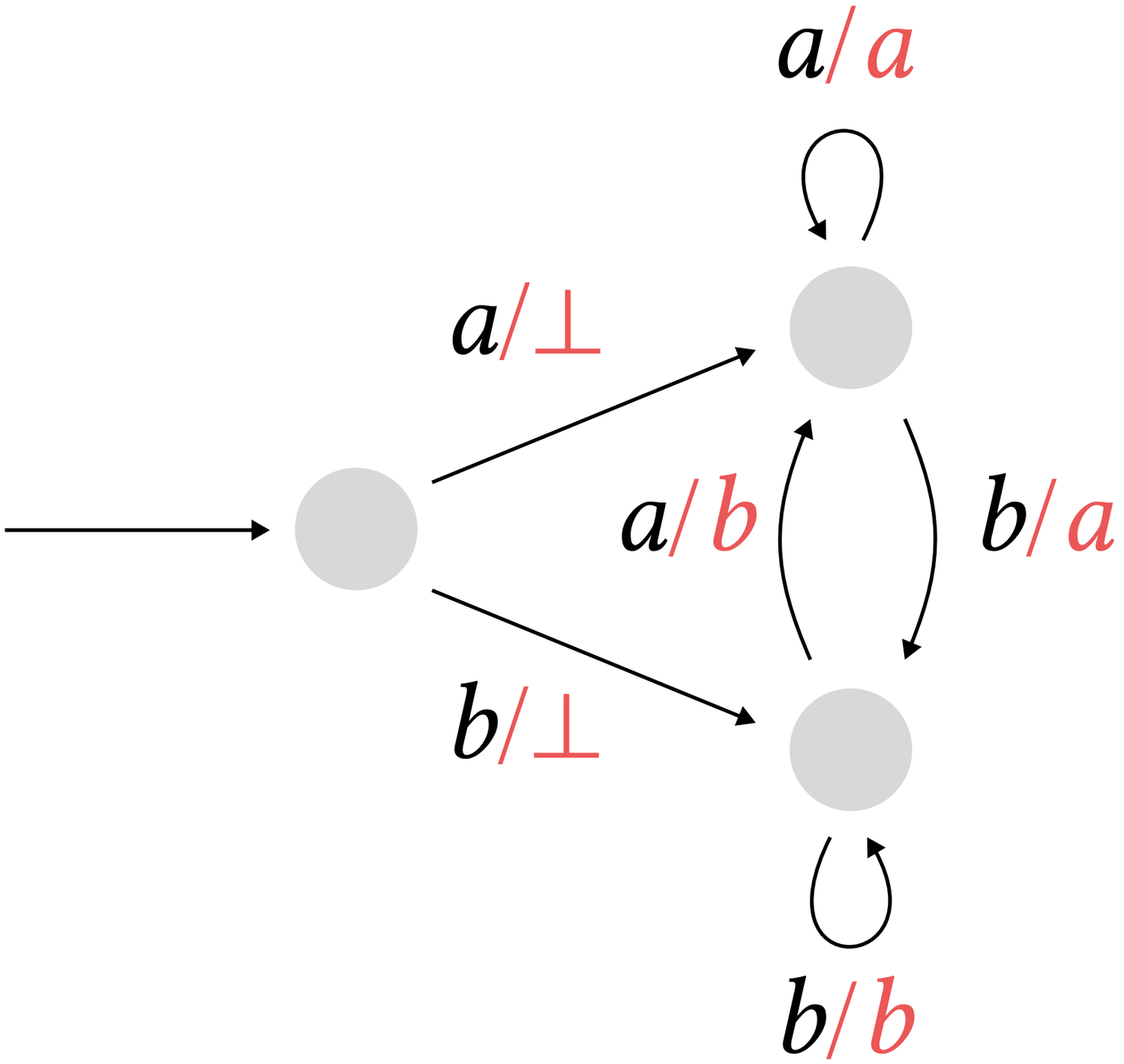

1 2 3 4 5 6 7 8 9 A B C D E F
b c a b c b a a c c b b a b c
input
δ11 δ12 δ44 δ45 δ46 δ99 δ9A δ9B δ9C δEE δEF
stage 1
δ13 δ37 δ88 δ9D
stage 2
δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13 δ13
stage 3

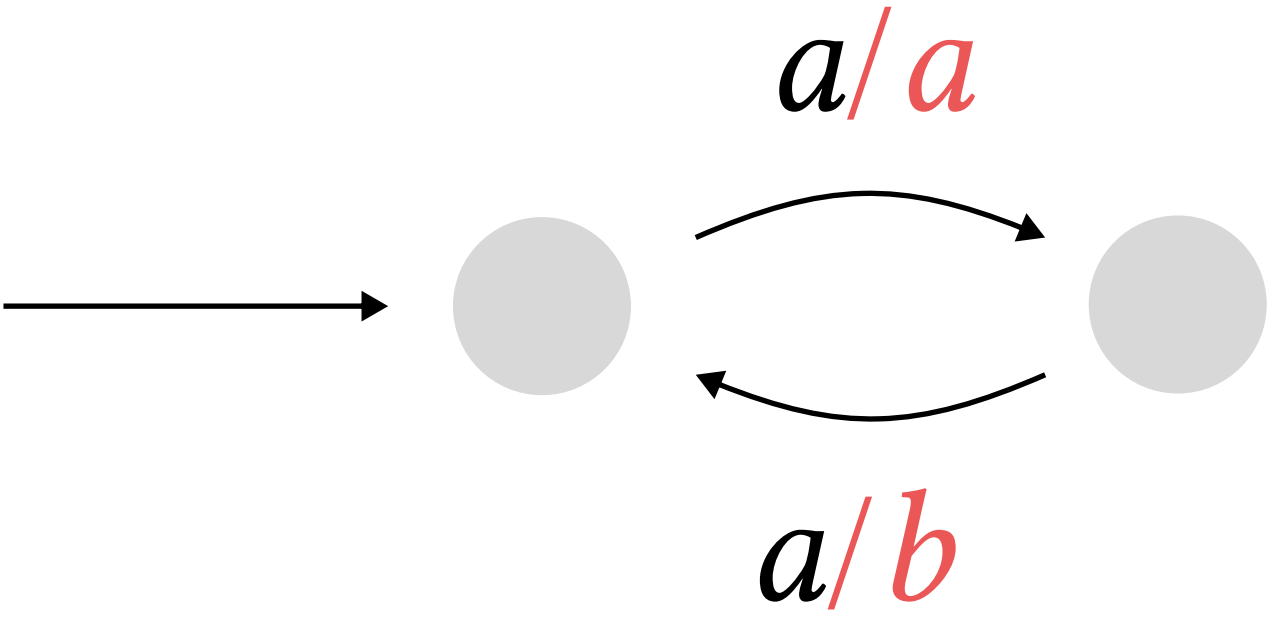

b c a b c b a a c b
input
⊥ b c a b c b a a c
output

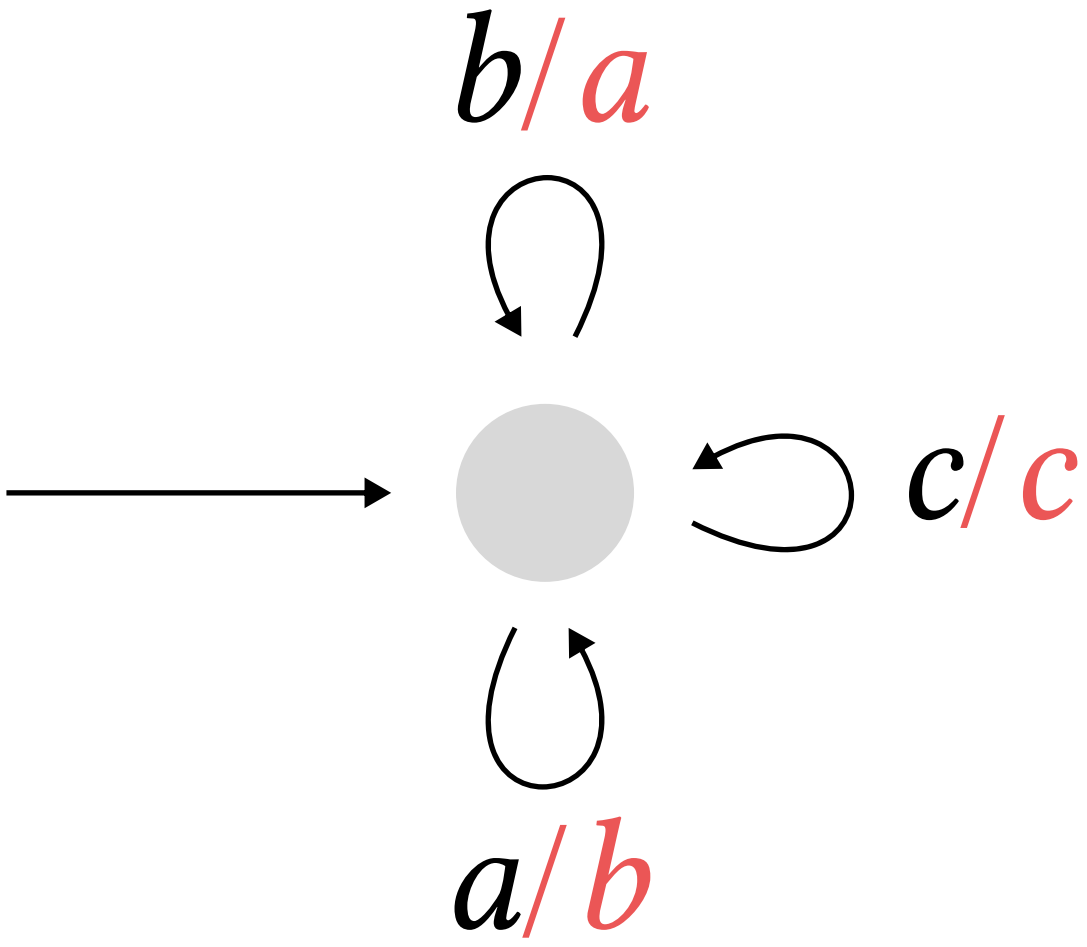

a a a a a a a a a a
input
a b a b a b a b a b
output


1 2 3 4 5 6 7 8 9 A B C D E F
b c # b c b # # c c b b # b c
input
Y11 Y12 Y13 Y14 Y15 Y16 Y17 Y18 Y19 Y1A Y1B Y1C Y1D Y1E Y1F
stage 1


1 2 3 4 5 6 7 8 9 A B C D E F
b c # b c b # # c c b b # b c
input
y_{11} y_{12} y_{13} y_{14} y_{15} y_{16} y_{17} y_{18} y_{19} y_{1A} y_{1B} y_{1C} y_{1D} y_{1E} y_{1F}
stage 1
y_{13} y_{13} y_{13} y_{13} y_{17} y_{18} y_{18} y_{18} y_{18} y_{18} y_{1D} y_{1D}
stage 2


a/a
a/⊥
a/b
b/a
b/⊥
b/b


a/a
a/b


b/a
c/c
a/b

a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8 a_9

q_0 q_1 q_2 q_3 q_4 q_5 q_6 q_7 q_8 q_9 run of f

b_1 b_2 b_3 b_4 b_5 b_6 b_7 b_8 b_9

p_0 p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8 p_9 run of g

c_1 c_2 c_3 c_4 c_5 c_6 c_7 c_8 c_9

input letter

a

source state

q

p

b

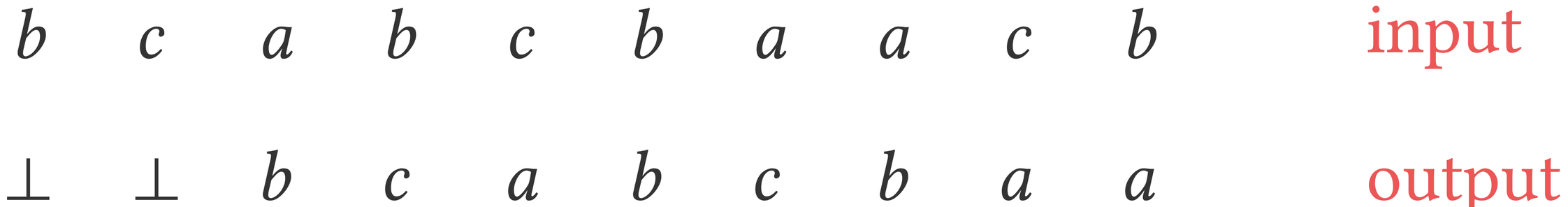
q’

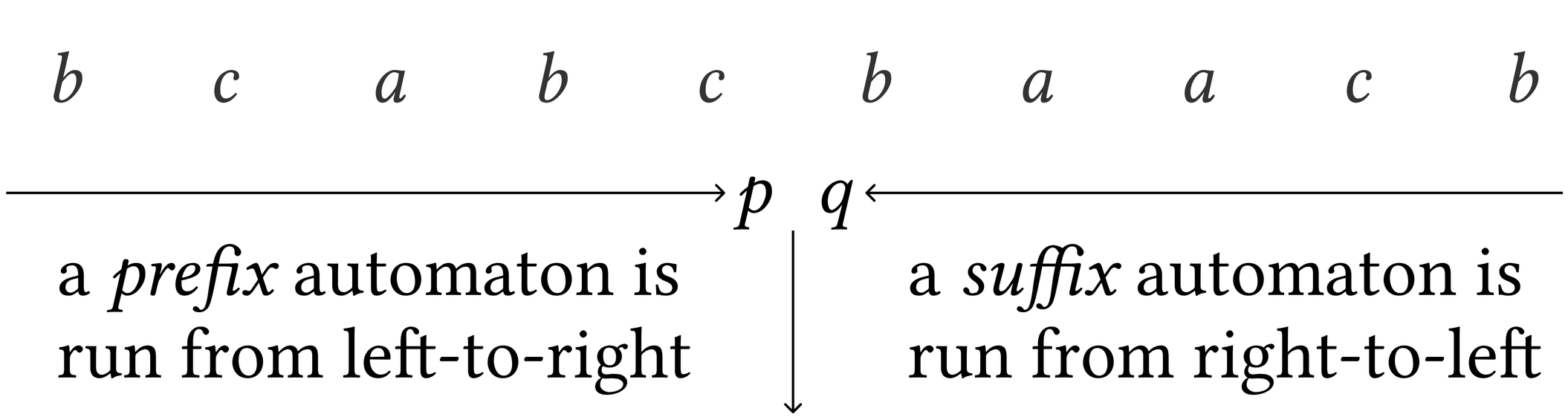
p’

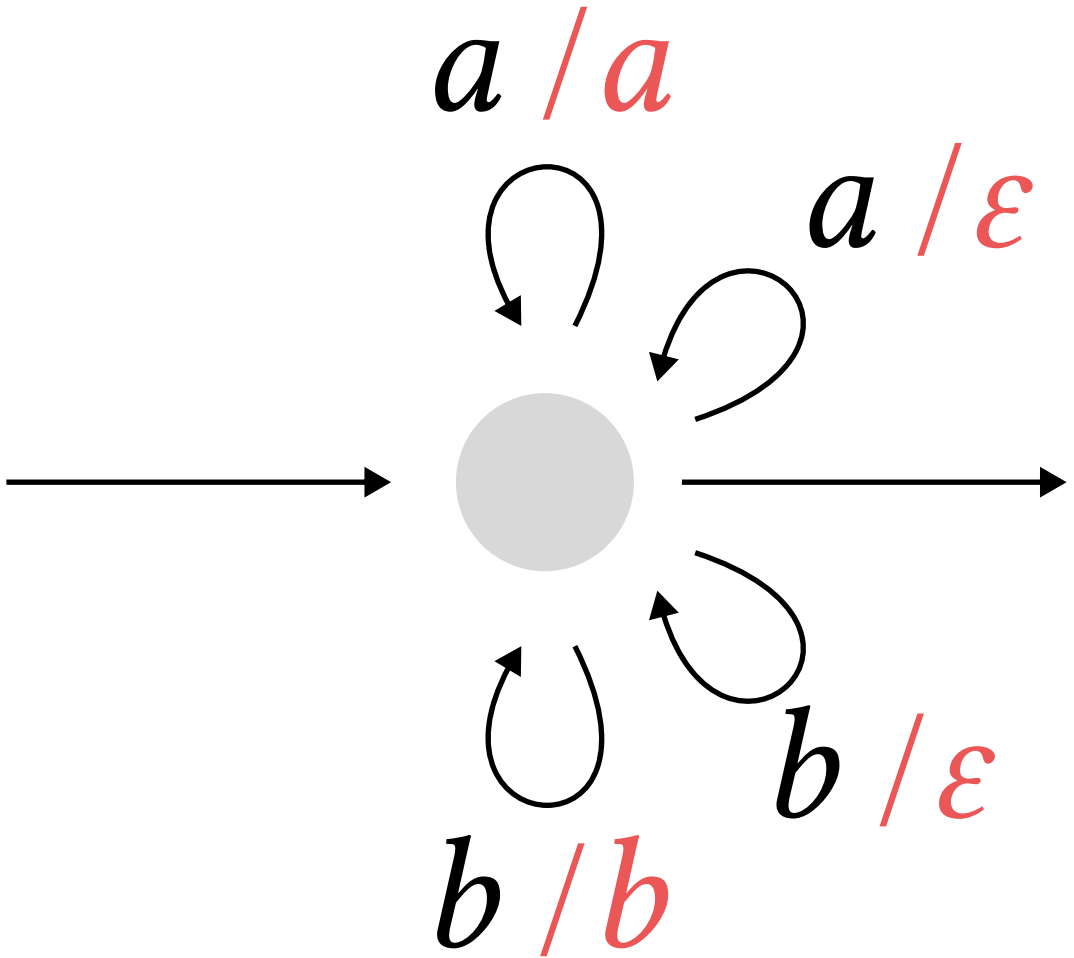
target state

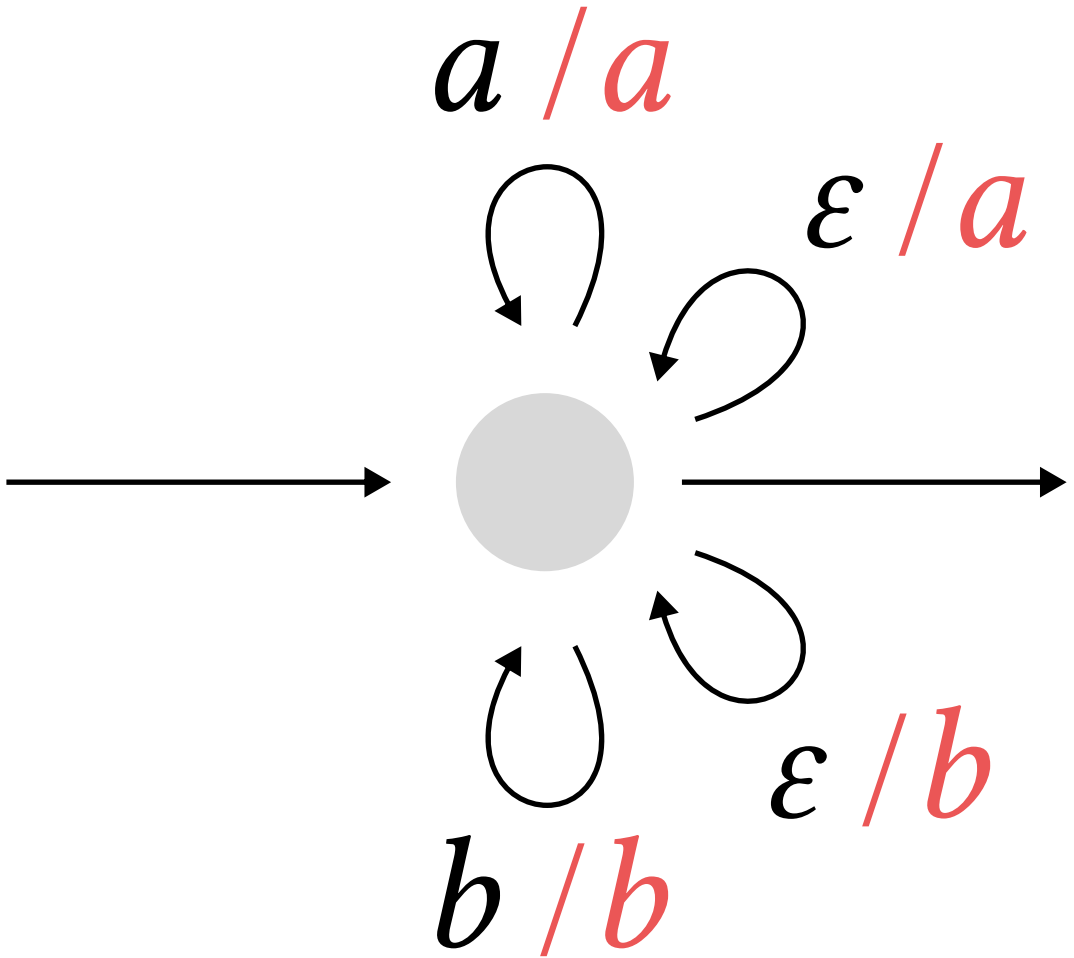
c

output letter

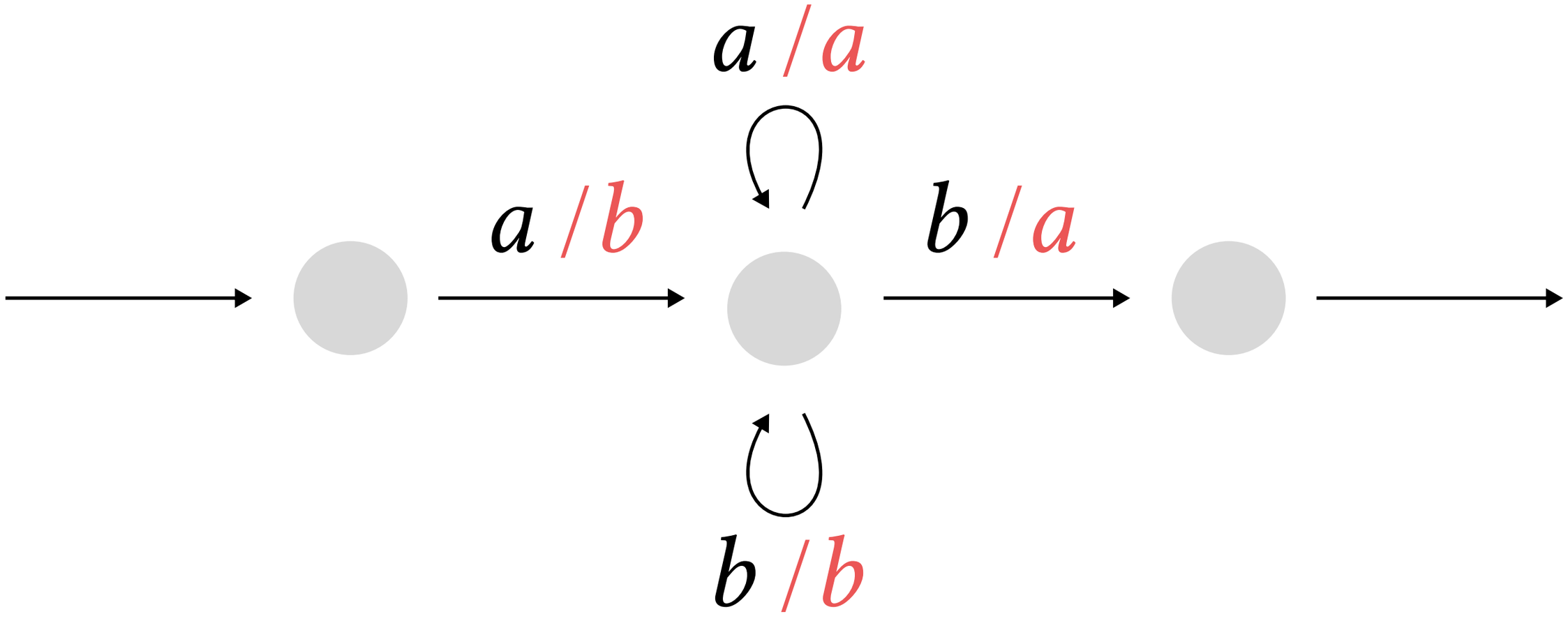

b c a b c b a a c b
input
⊥ ⊥ b c a b c b a a
output


b c a b c b a a c b
p q
a *prefix* automaton is run from left-to-right
a *suffix* automaton is run from right-to-left
cdd
based on the states of the prefix and suffix automata, a piece of the output is produced


a / a
a / ε
b / ε
b / b


a / a
ε / a
ε / b
b / b

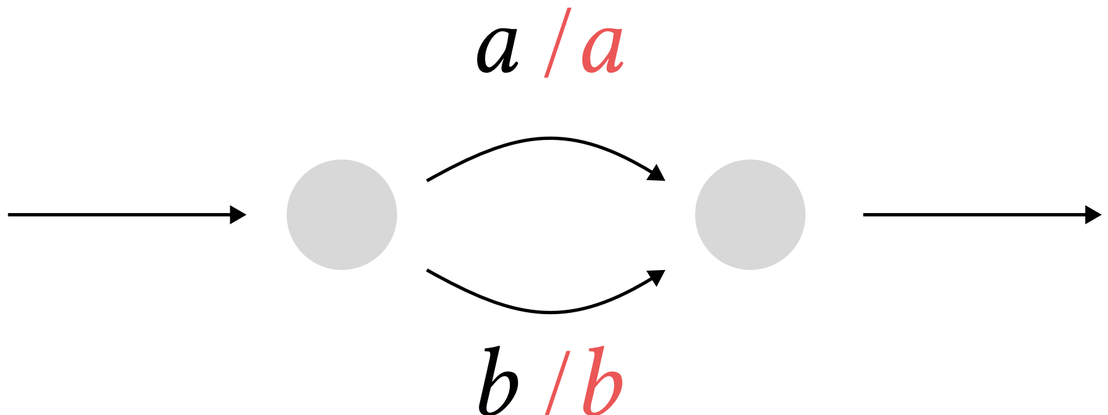
inputs that start with f and end with b

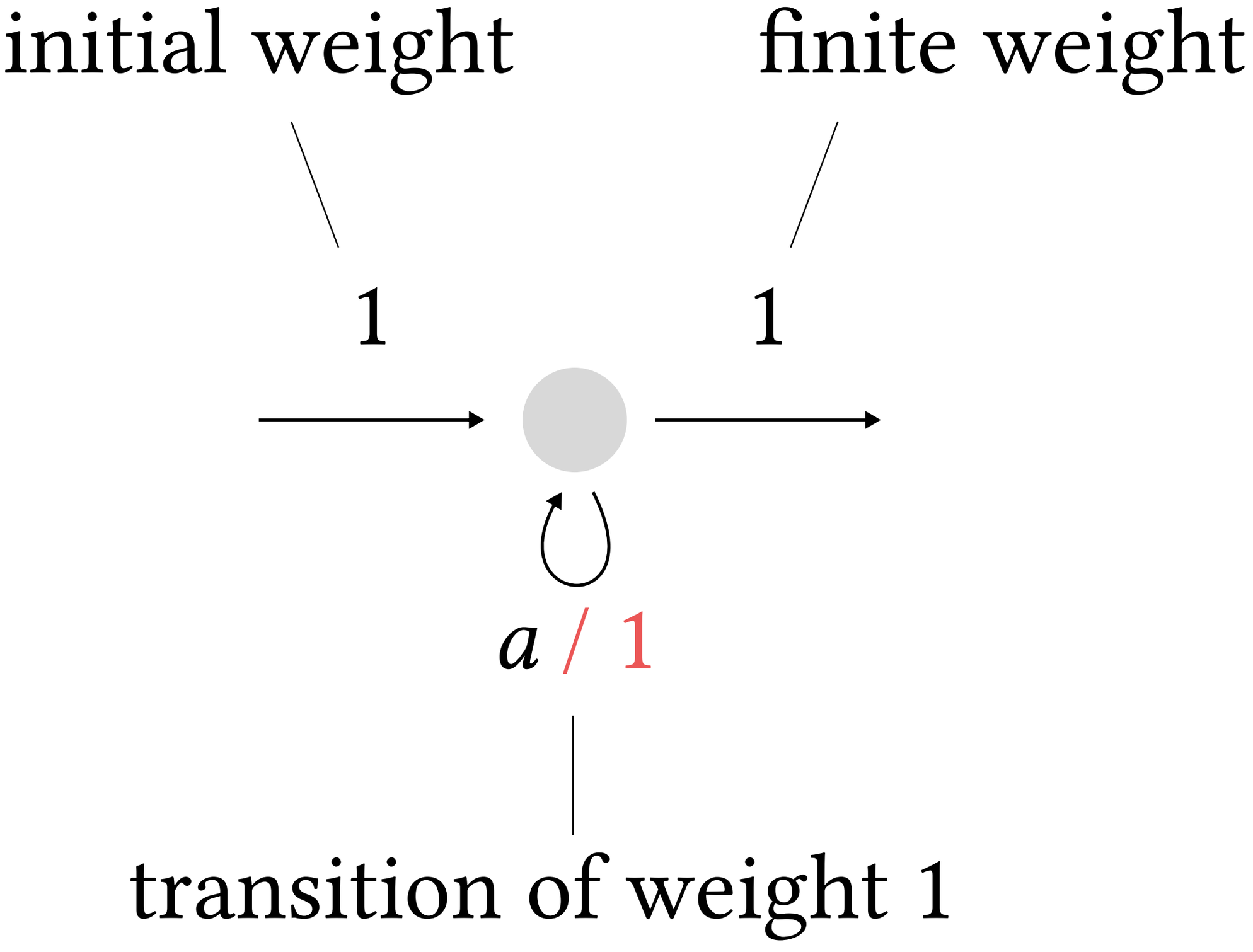


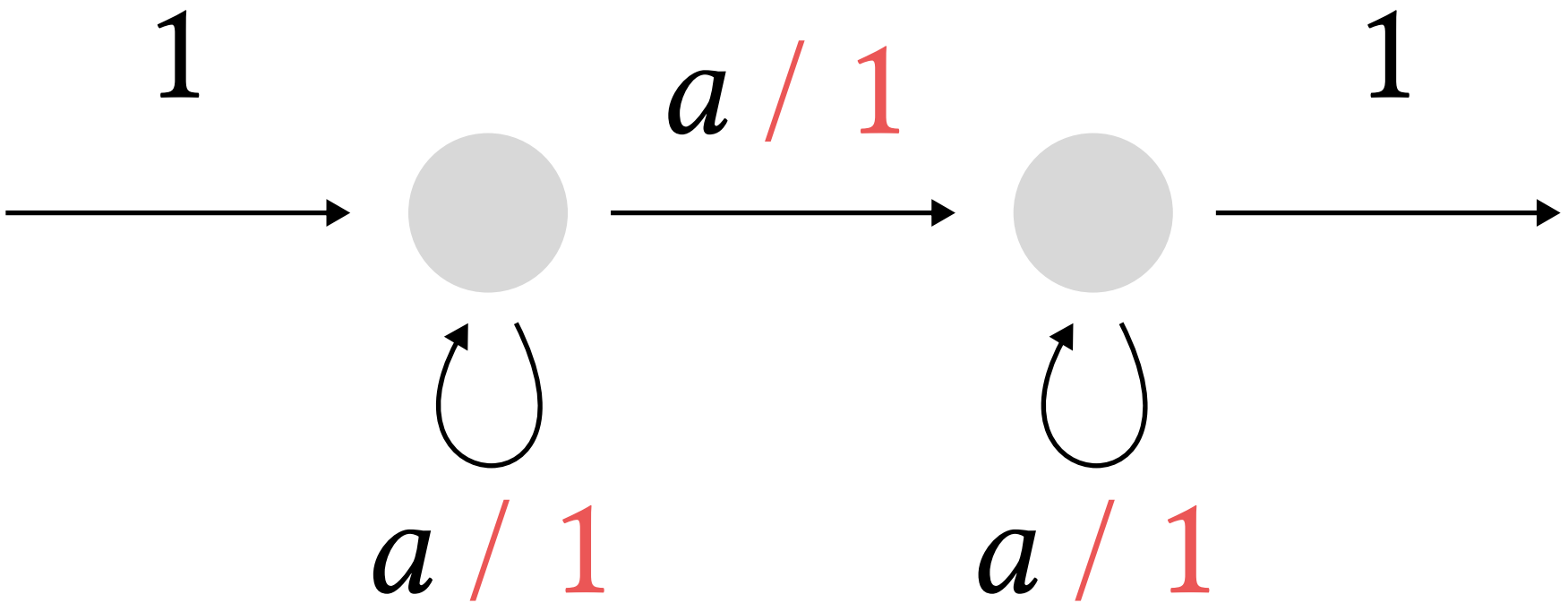
empty inputs

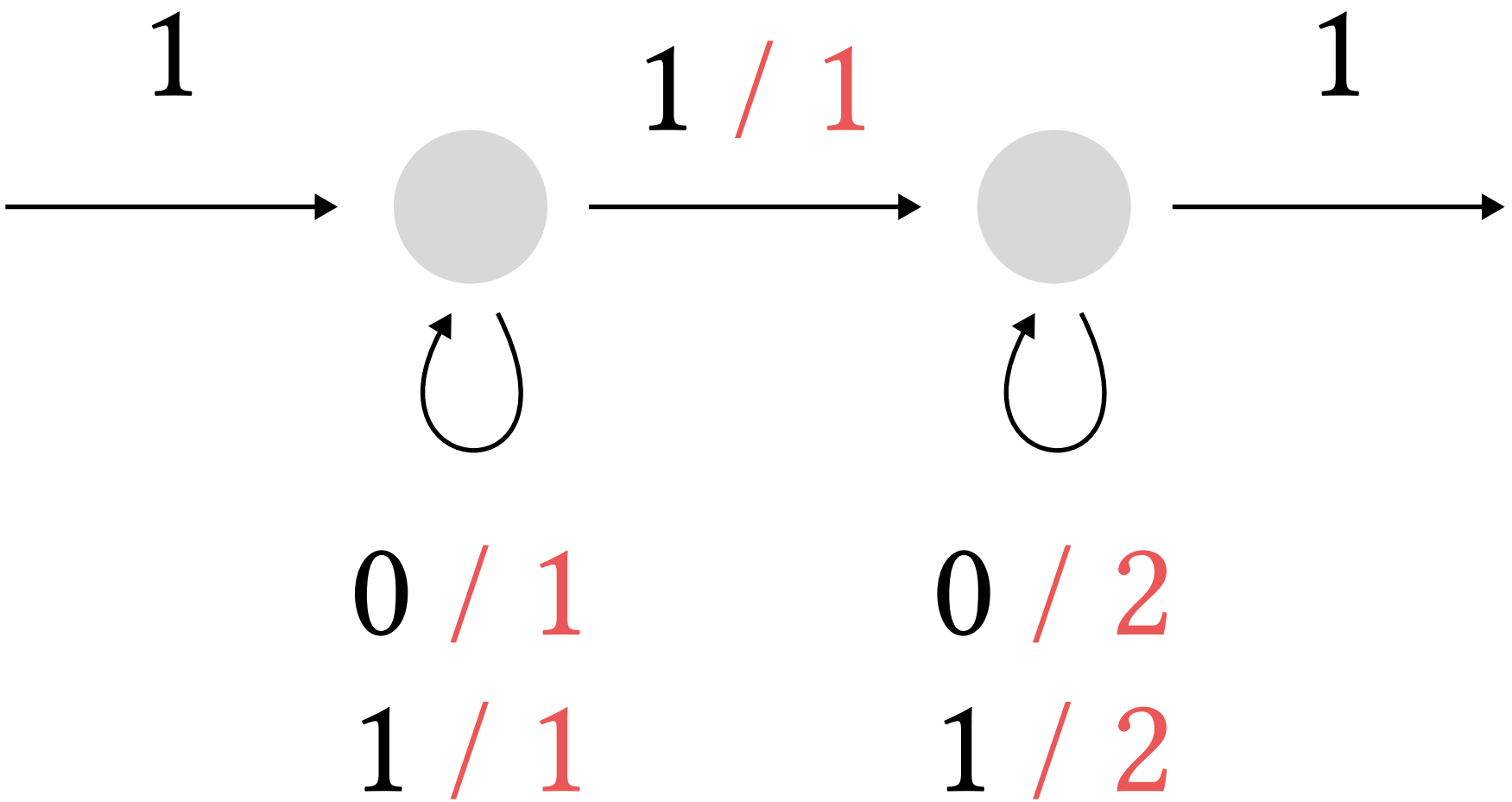


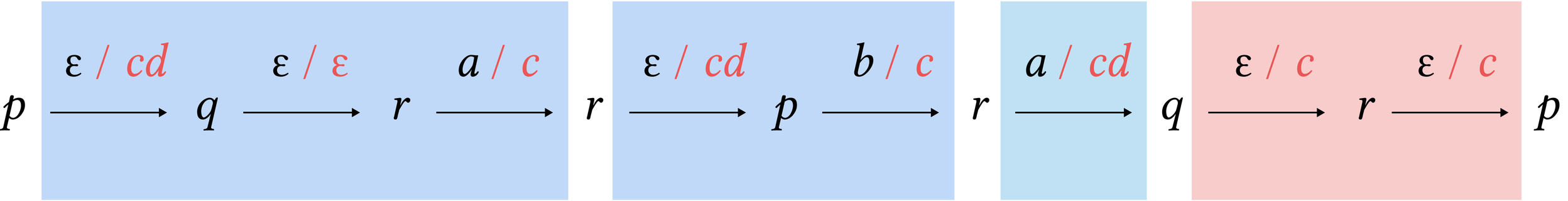
one letter inputs

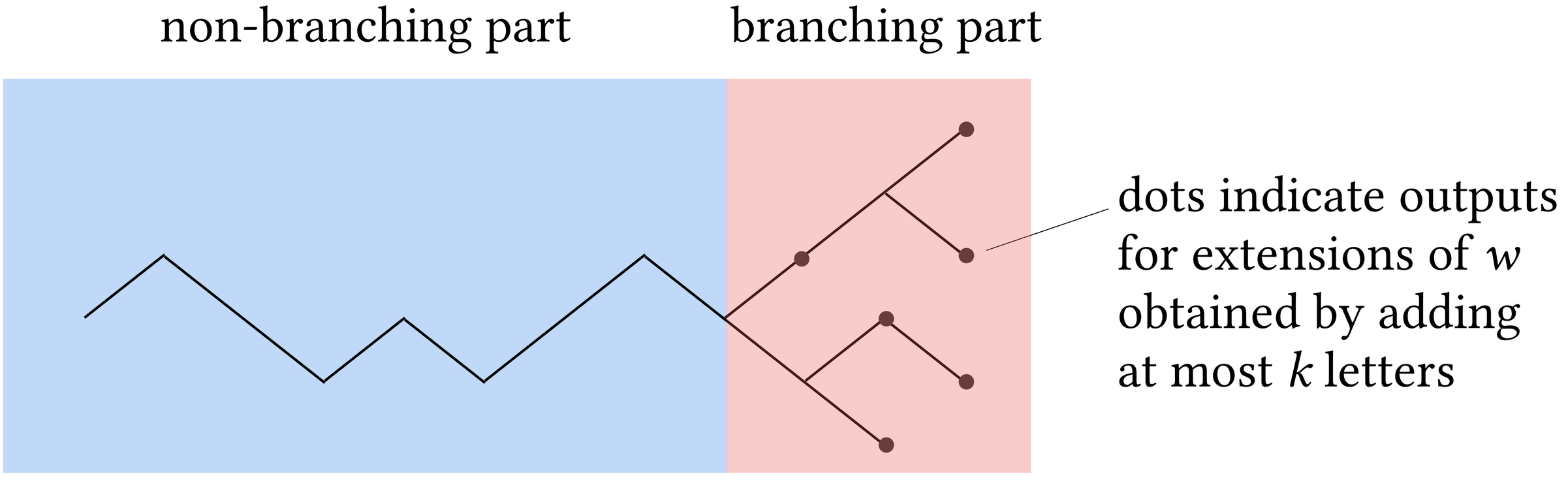


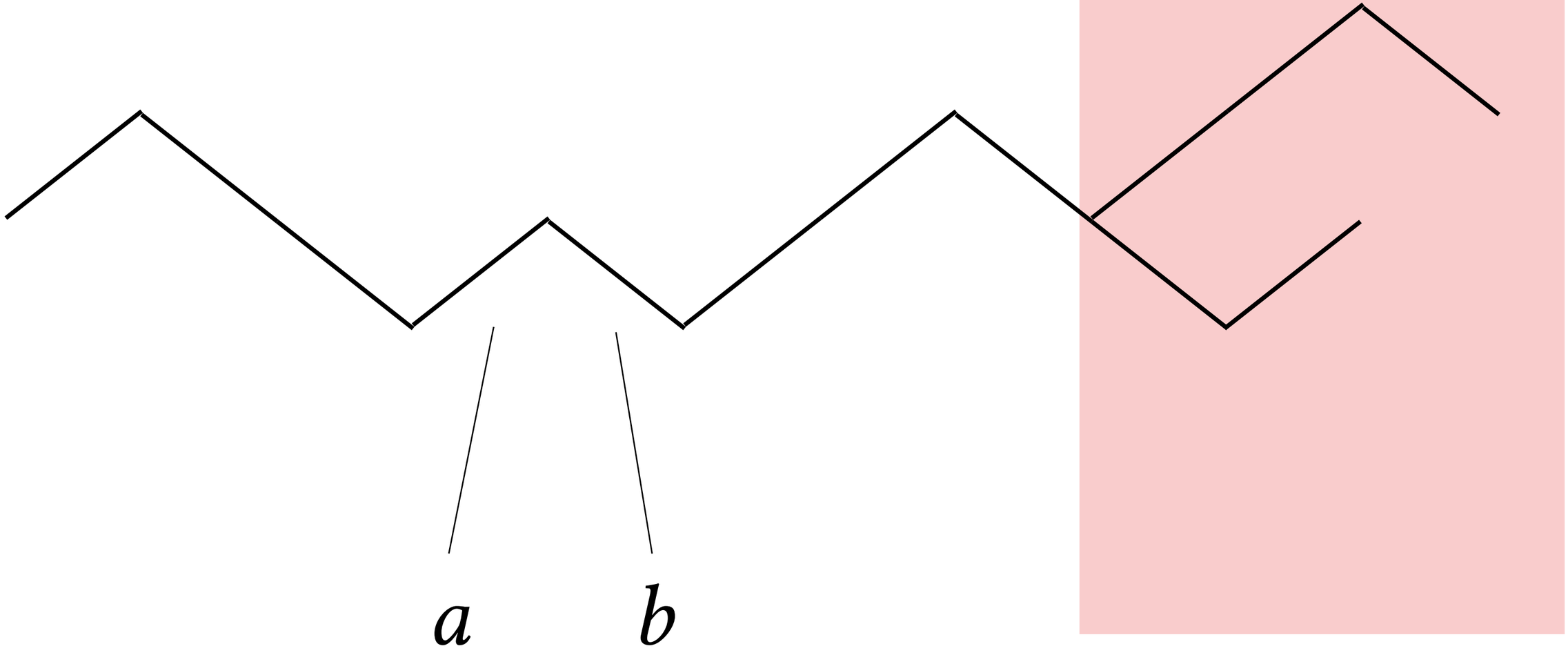

initial weight
finite weight
1
1
a / 1
transition of weight 1


1
a / 1
1
a / 1
a / 1


1
1 / 1
1
0 / 1
1 / 1
0 / 2
1 / 2

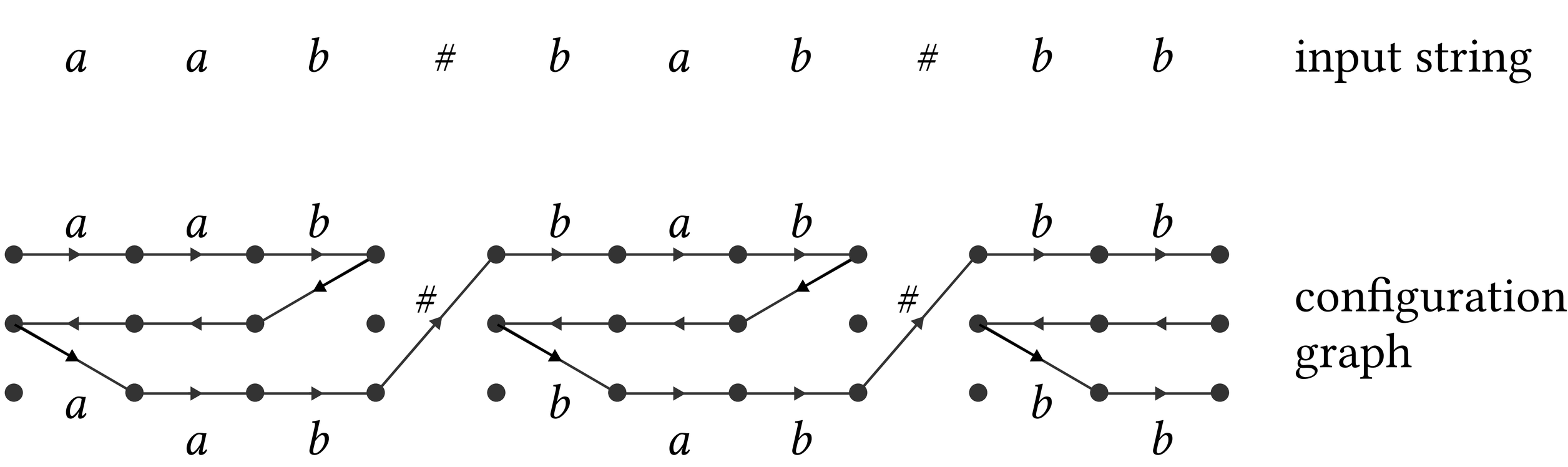

ε / cd
ε / ε
a / c
ε / cd
b / c
a / cd
ε / c
ε / c
p
q
r
r
p
r
q
r
p

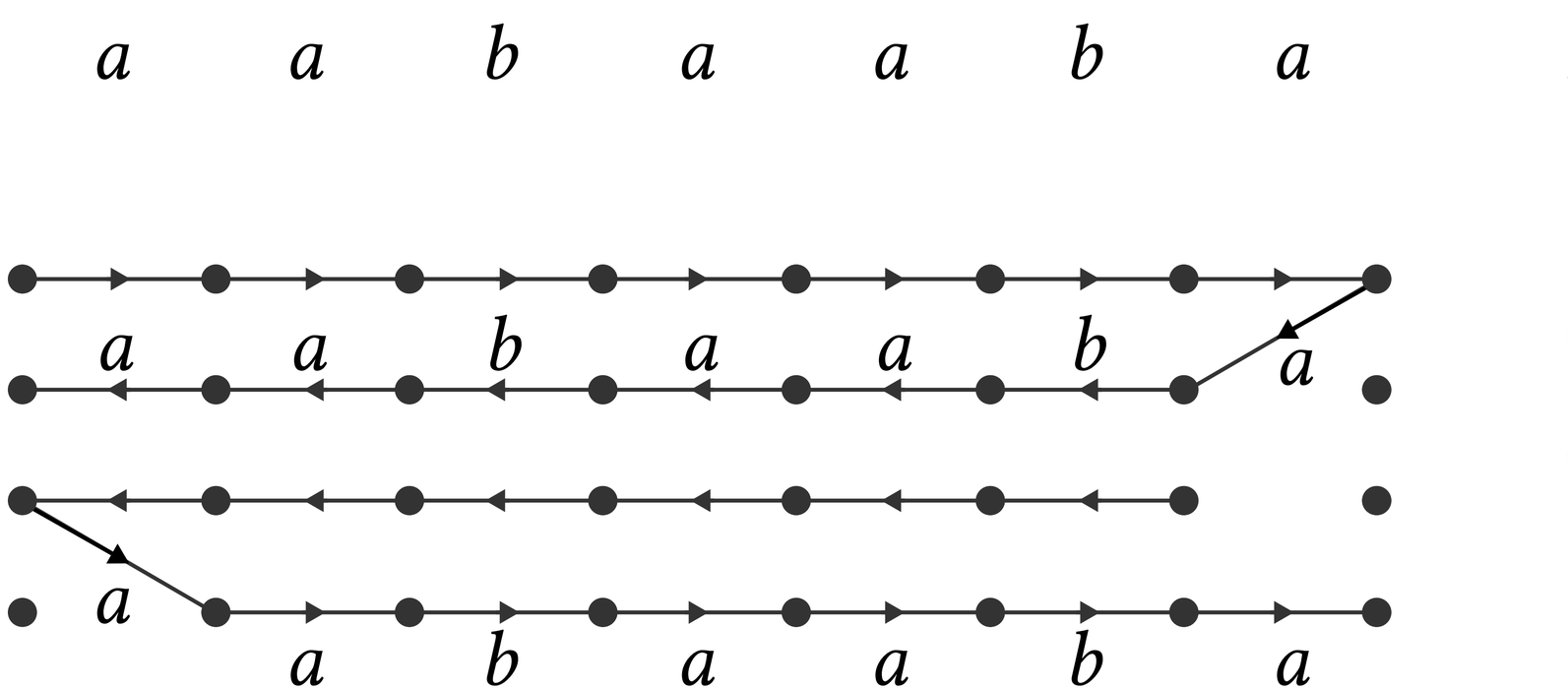

non-branching part
branching part
dots indicate outputs
for extensions of w
obtained by adding
at most k letters

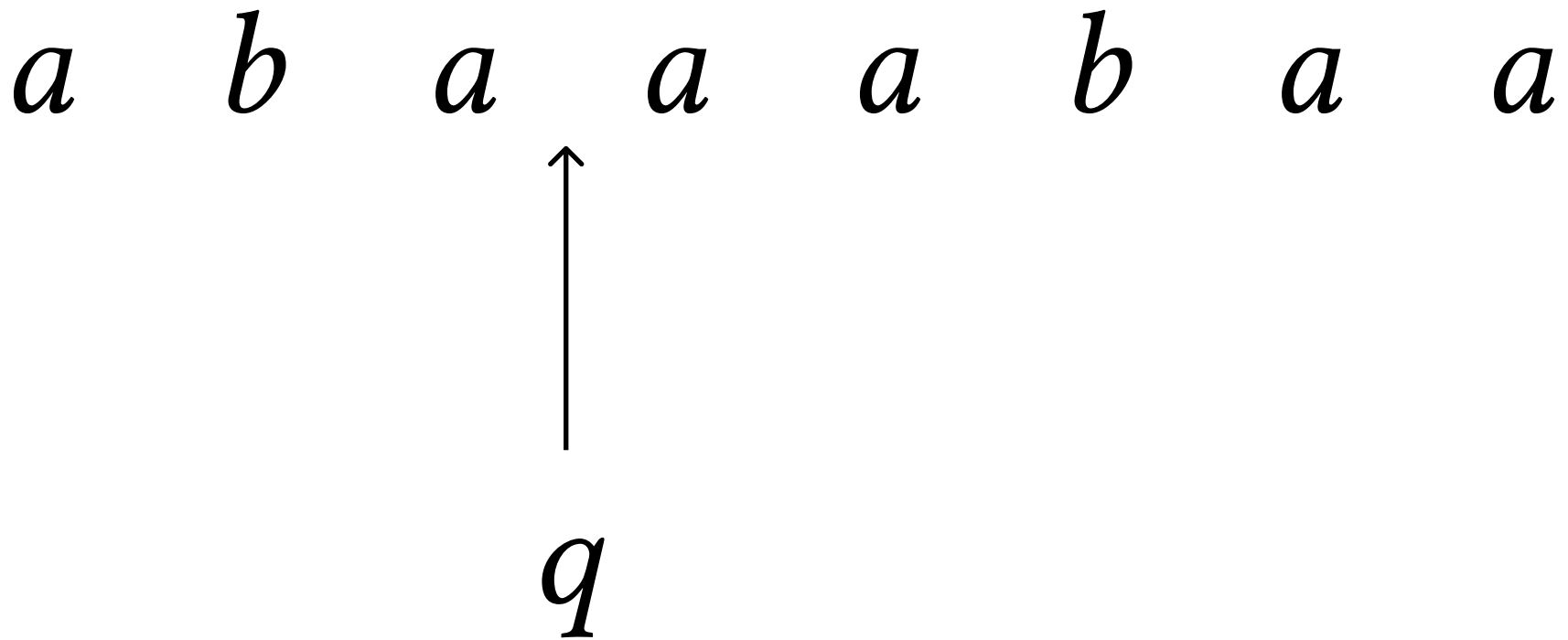


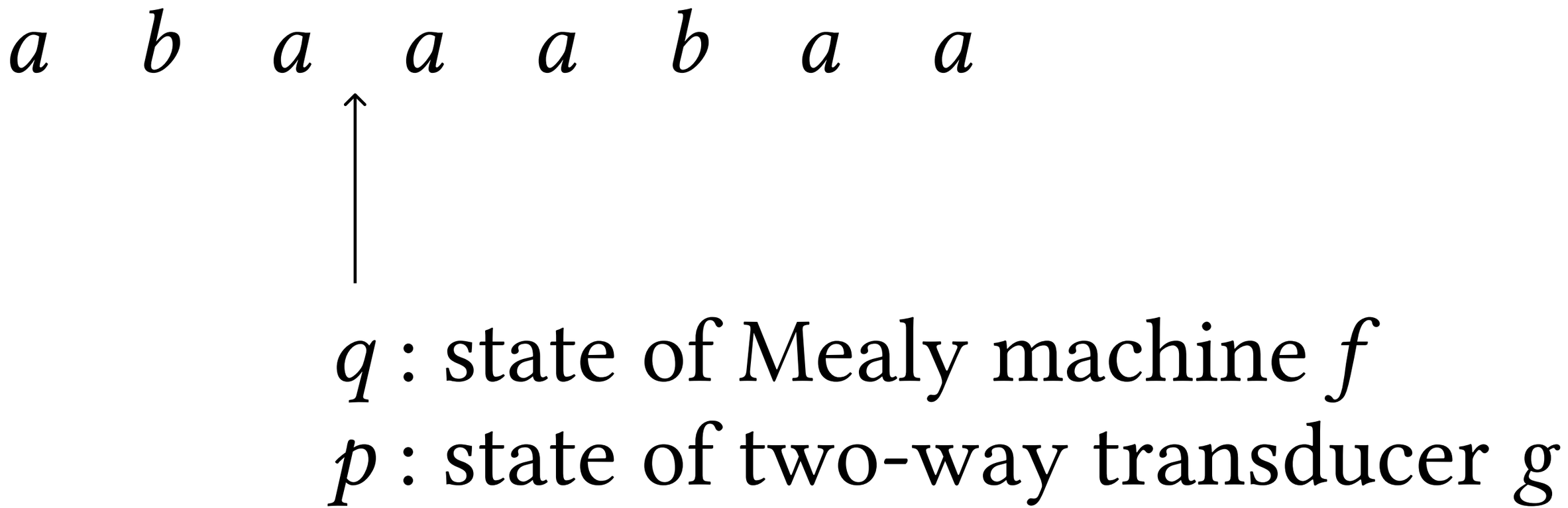
directions indicate letters

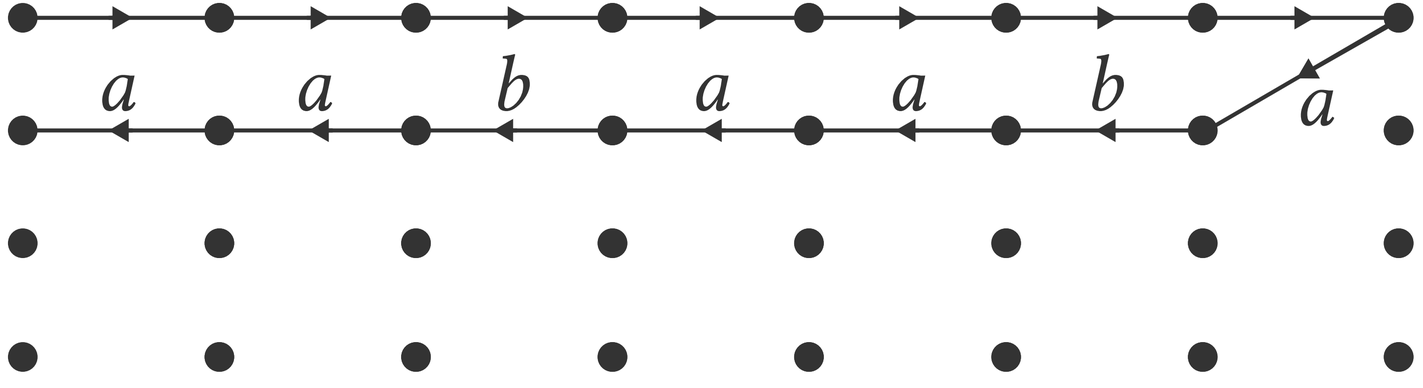
left distance $k = 3$


a a b # b a b # b b
input string
a a b # b a b # b b
a a b b a b b b
configuration graph

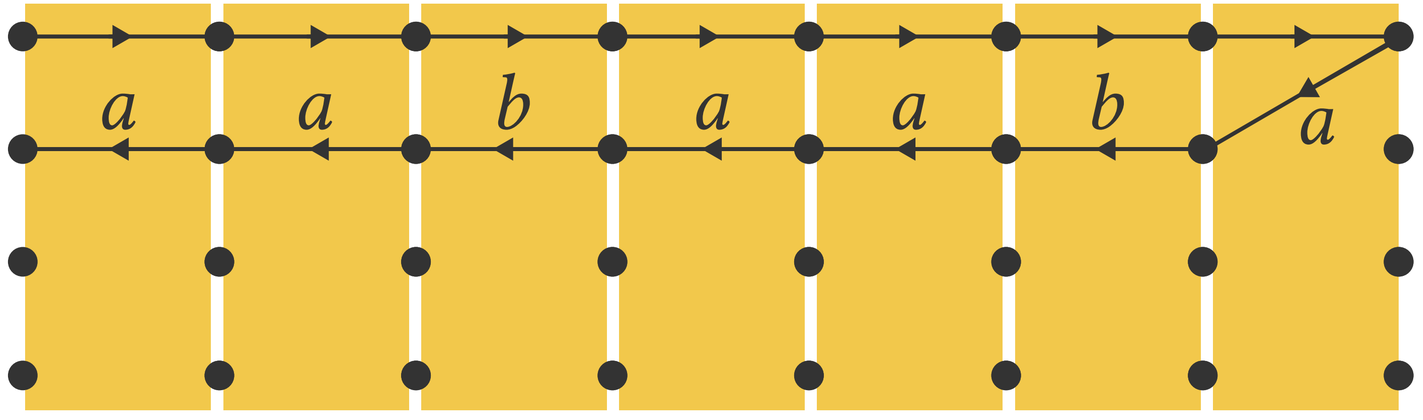

a a b a a b a
input string
a a b a a b a
a
a b a a b a
configuration
graph


a b a a a b a a
q


a b a a a b a a
q : state of Mealy machine f
p : state of two-way transducer g

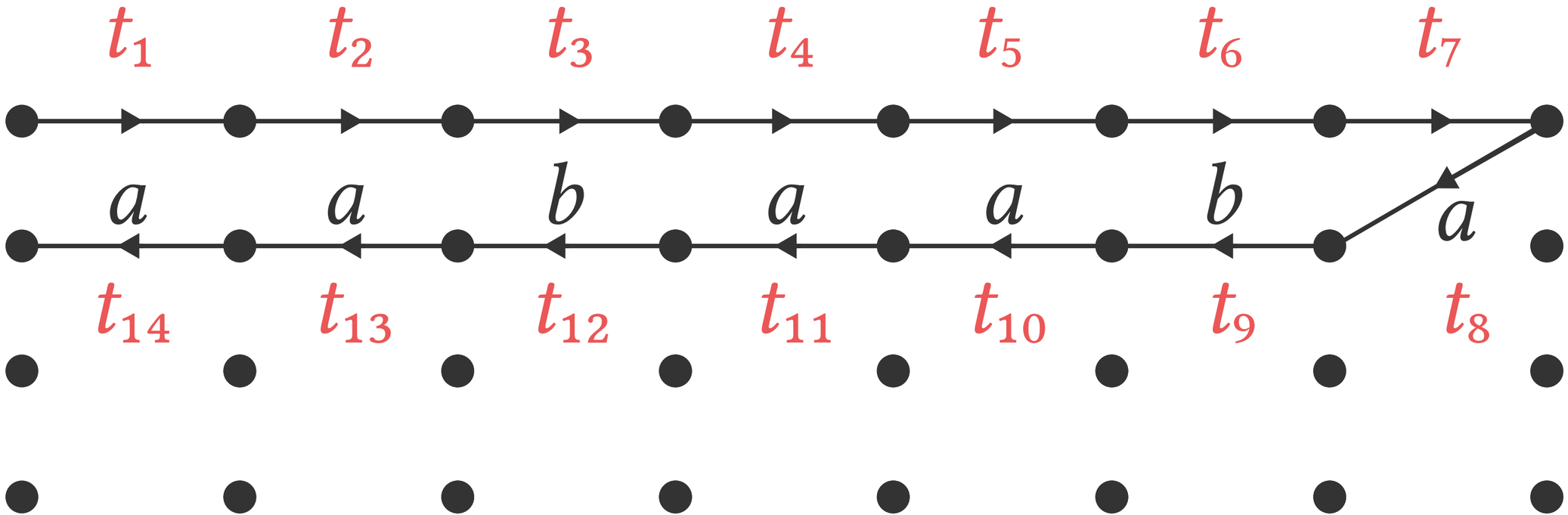


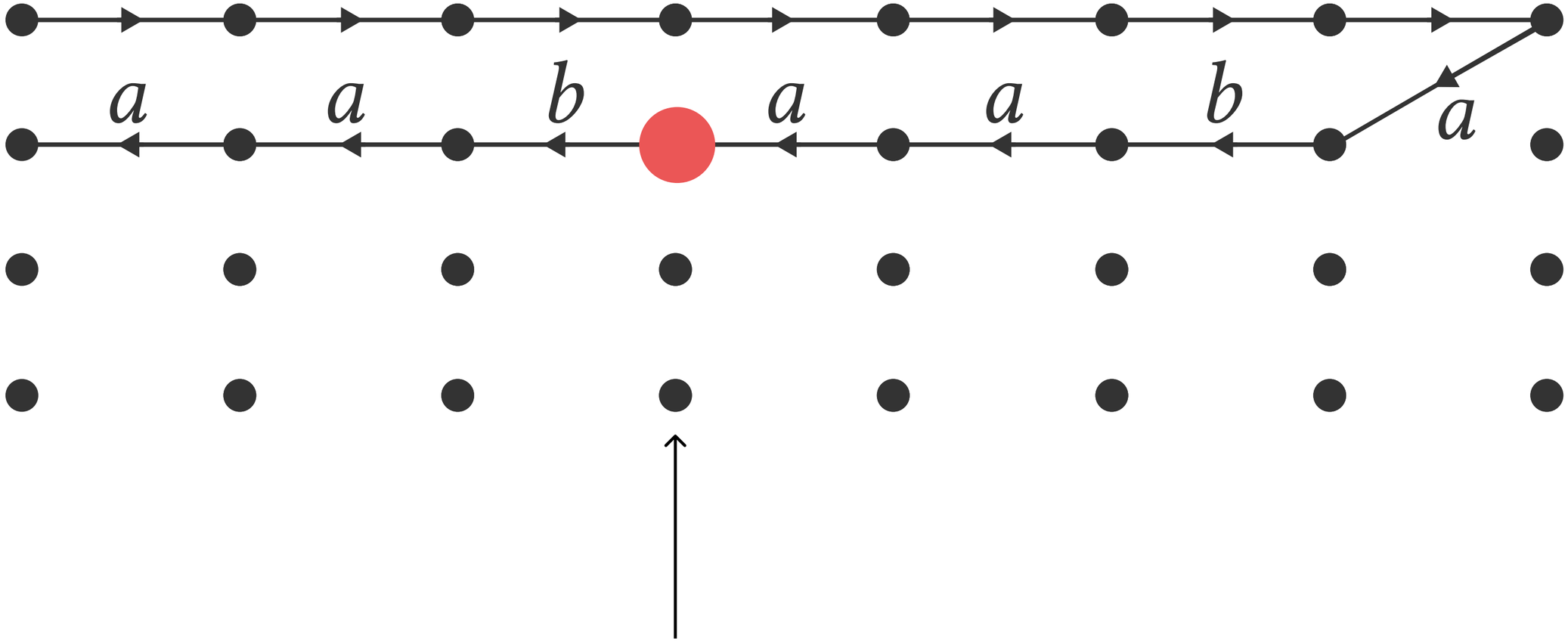
reachable part of the configuration graph



string
representation

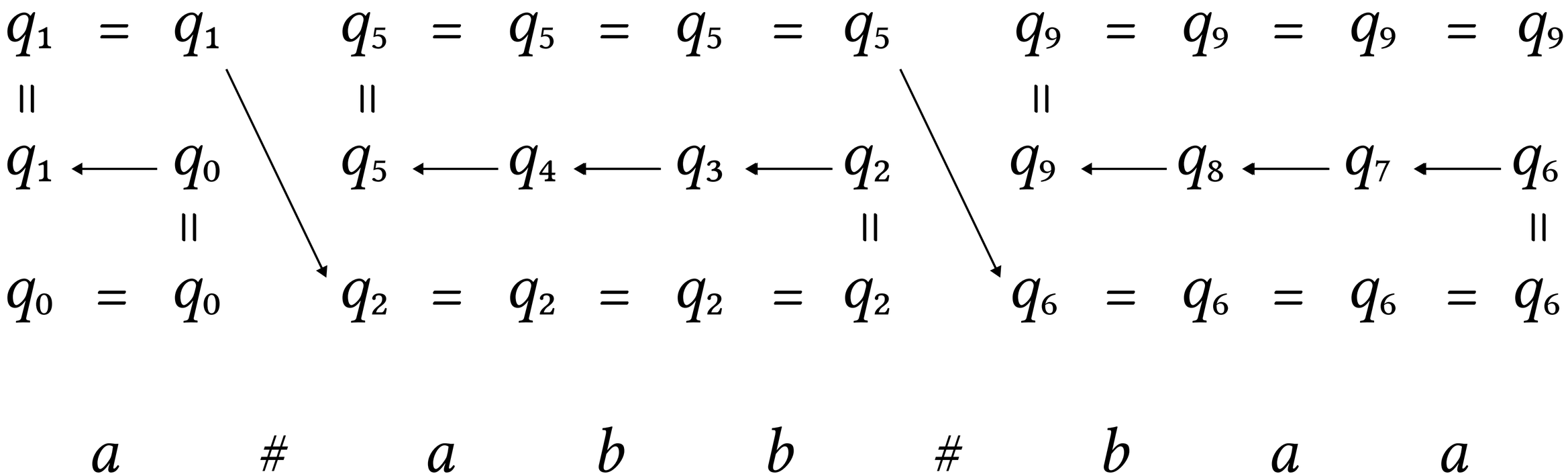


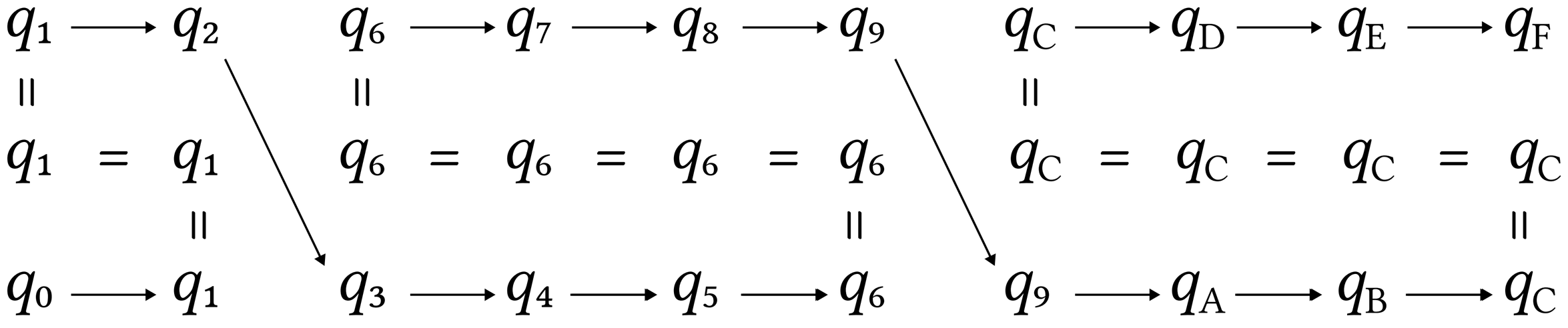

t_1 t_2 t_3 t_4 t_5 t_6 t_7
a a b a a b a
t_{14} t_{13} t_{12} t_{11} t_{10} t_9 t_8

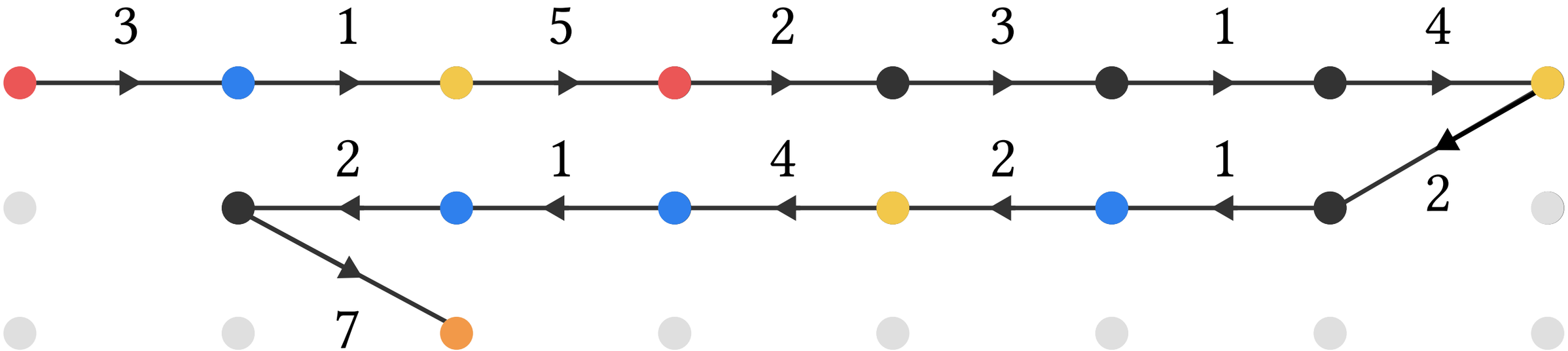


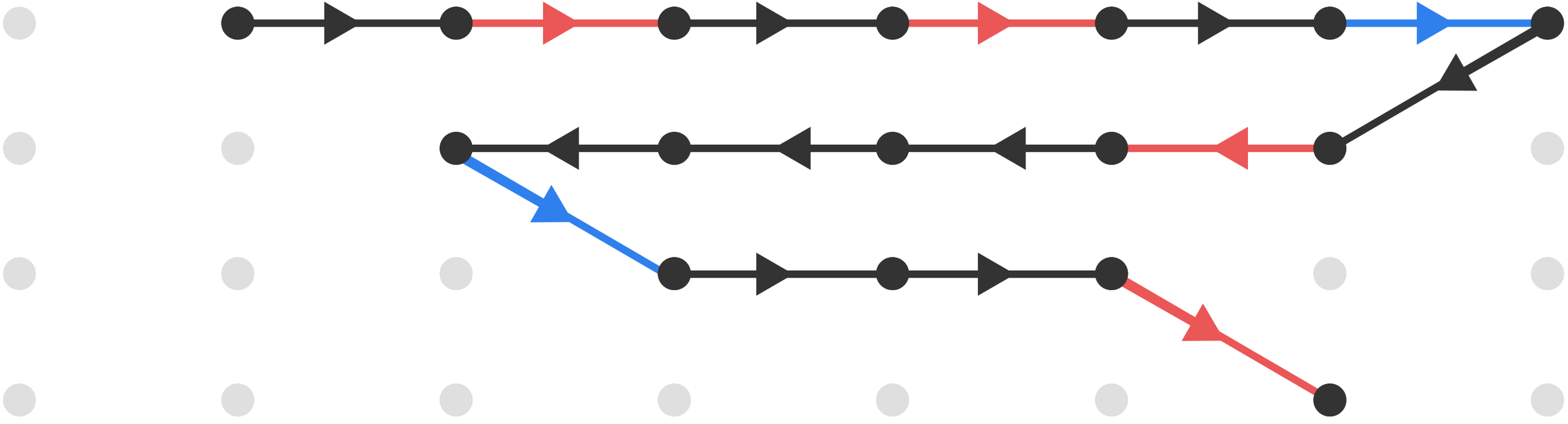
product automaton remembers:

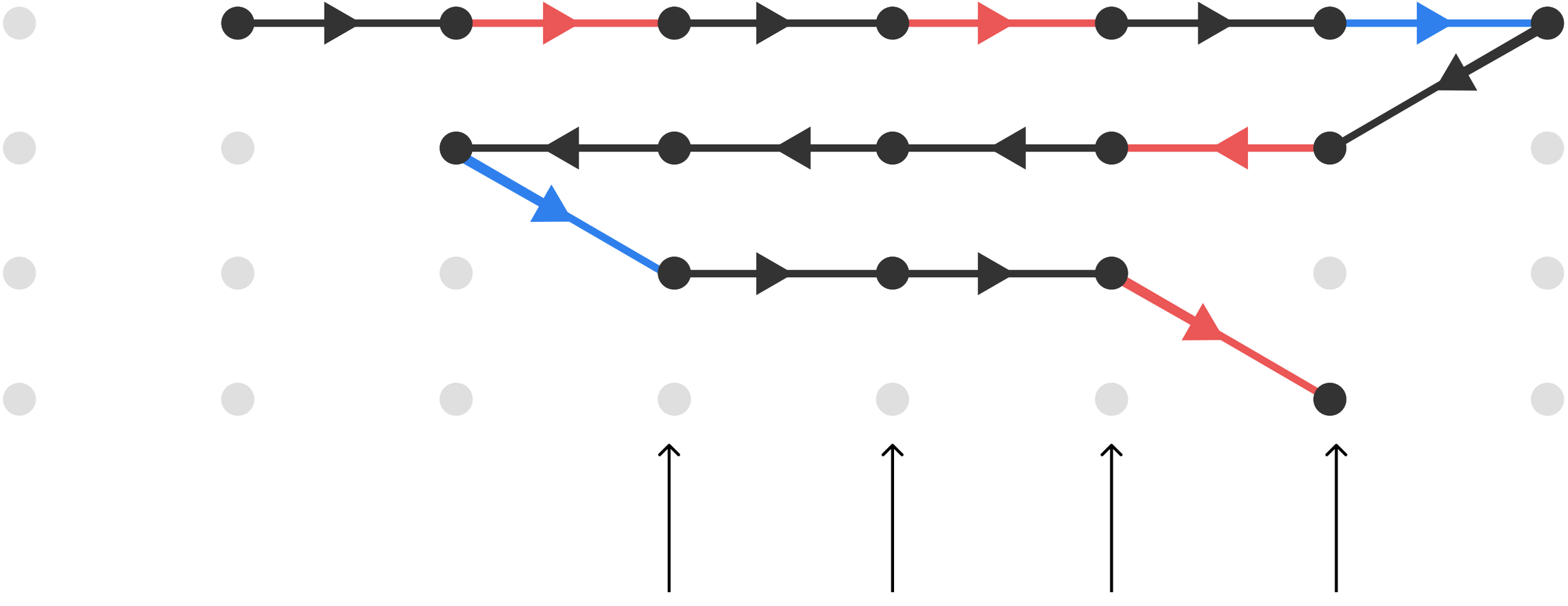
- state of f (red circle); and
- state of g

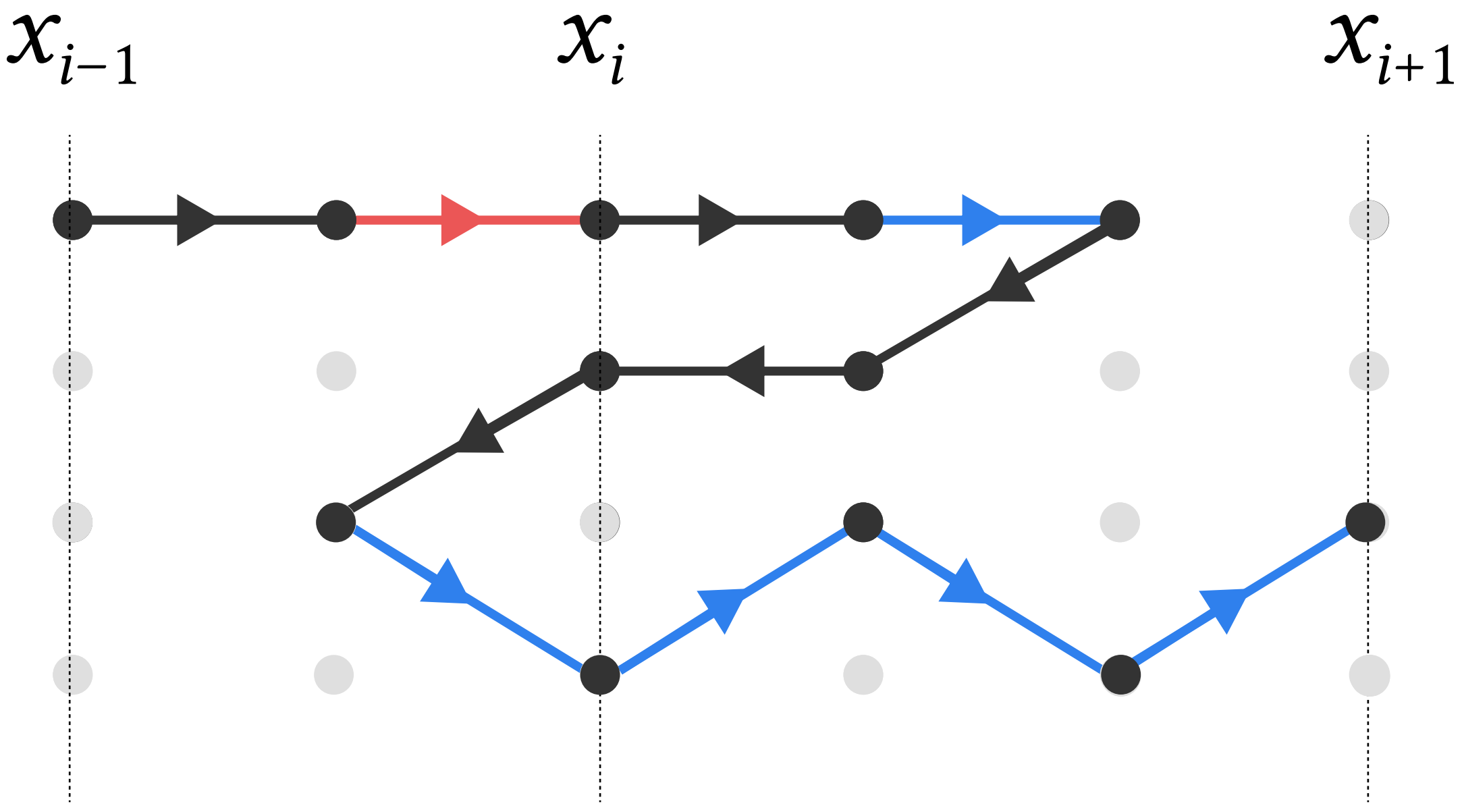

$q_1 = q_1$ $q_5 = q_5 = q_5 = q_5$ $q_9 = q_9 = q_9 = q_9$
$q_1 \leftarrow q_0$ $q_5 \leftarrow q_4 \leftarrow q_3 \leftarrow q_2$ $q_9 \leftarrow q_8 \leftarrow q_7 \leftarrow q_6$
$q_0 = q_0$ $q_2 = q_2 = q_2 = q_2$ $q_6 = q_6 = q_6 = q_6$
a $\#$ a b b $\#$ b a a

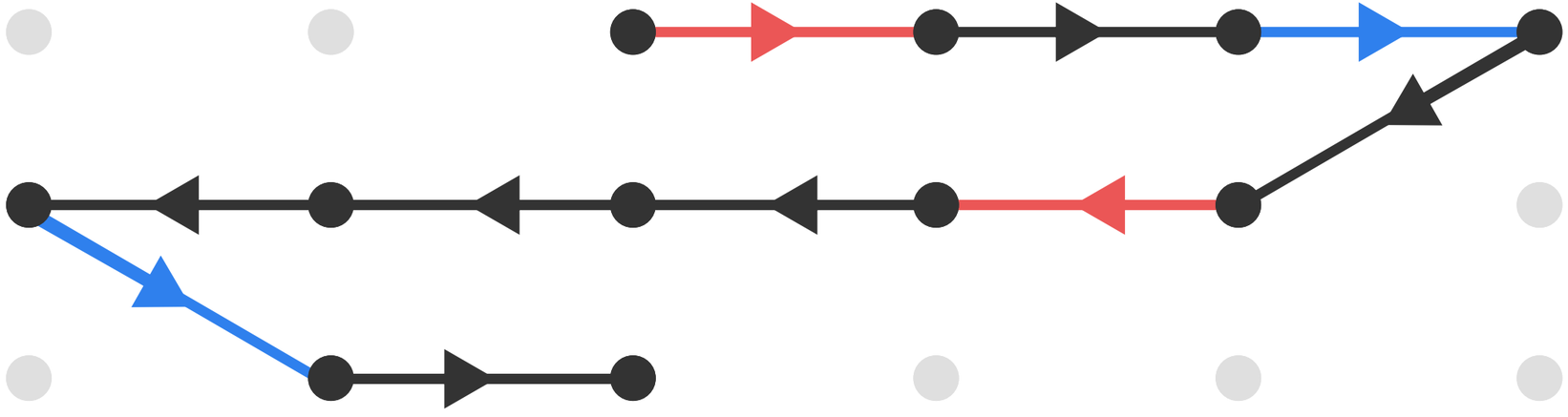

$q_1 \longrightarrow q_2$
$q_6 \longrightarrow q_7 \longrightarrow q_8 \longrightarrow q_9$
$q_C \longrightarrow q_D \longrightarrow q_E \longrightarrow q_F$
$q_1 = q_1$
$q_6 = q_6 = q_6 = q_6$
$q_C = q_C = q_C = q_C$
$q_0 \longrightarrow q_1$
$q_3 \longrightarrow q_4 \longrightarrow q_5 \longrightarrow q_6$
$q_9 \longrightarrow q_A \longrightarrow q_B \longrightarrow q_C$

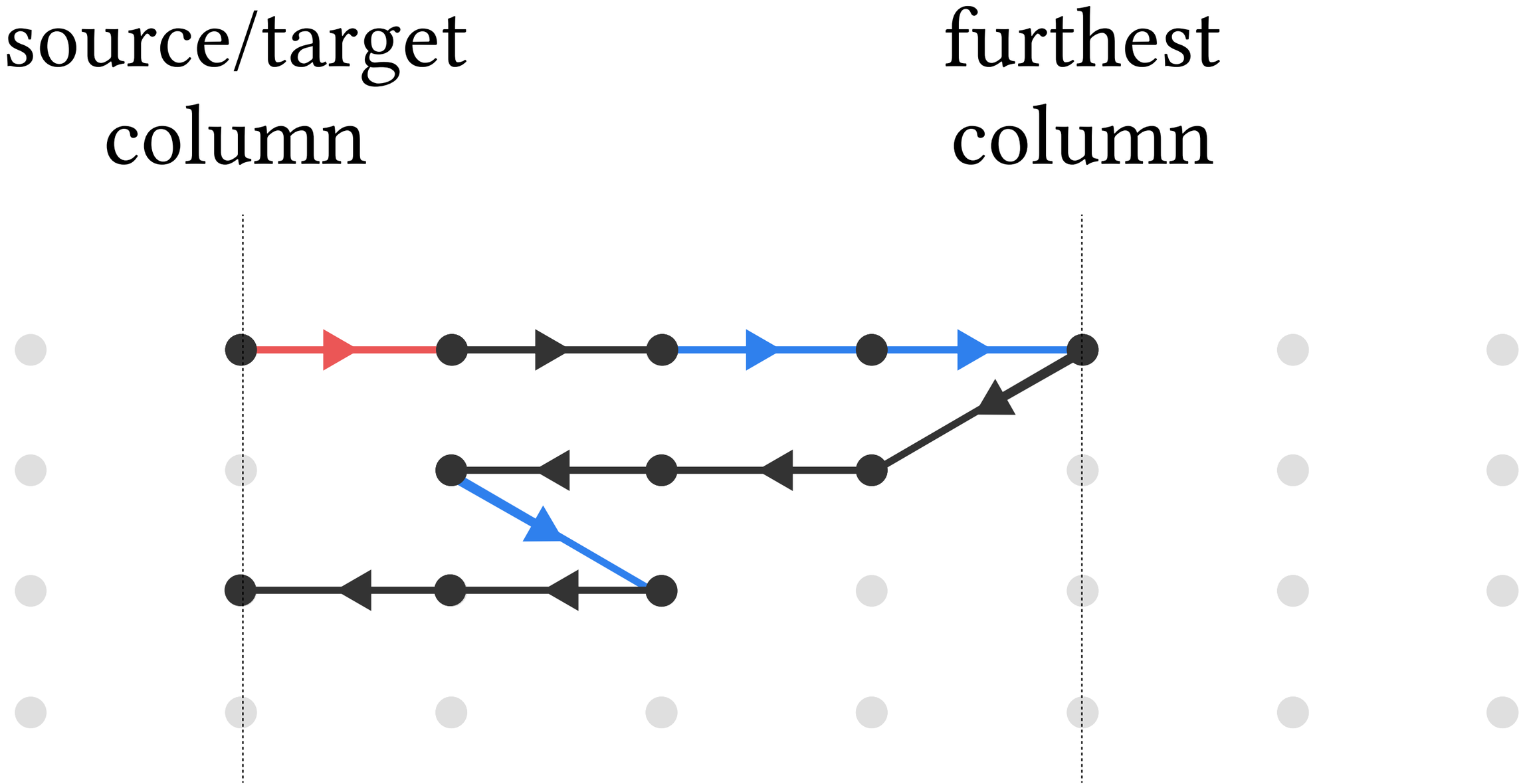

3
1
5
2
3
1
4
2
1
4
2
1
2
7

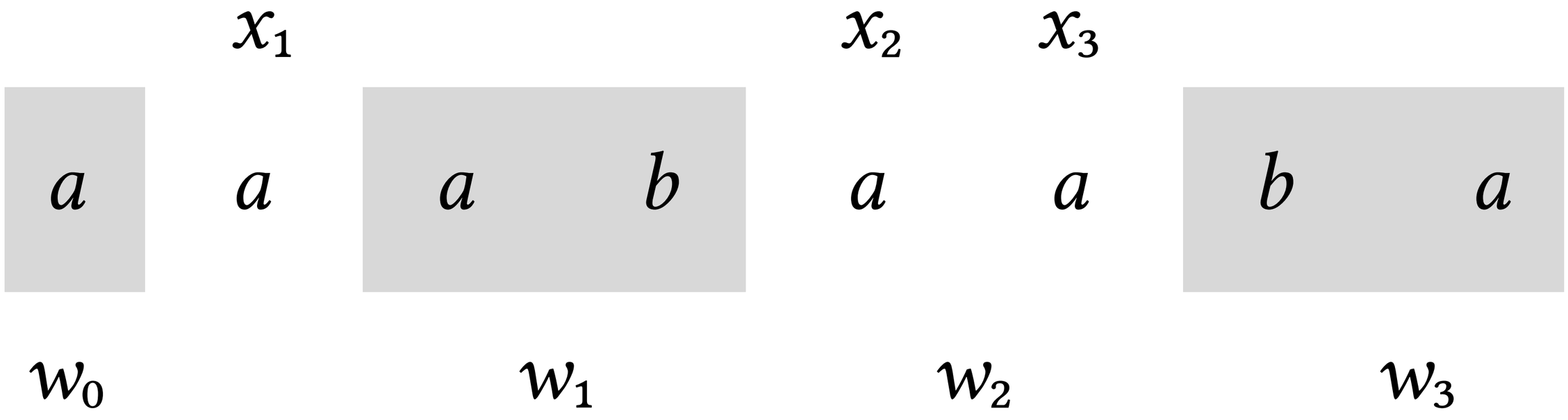


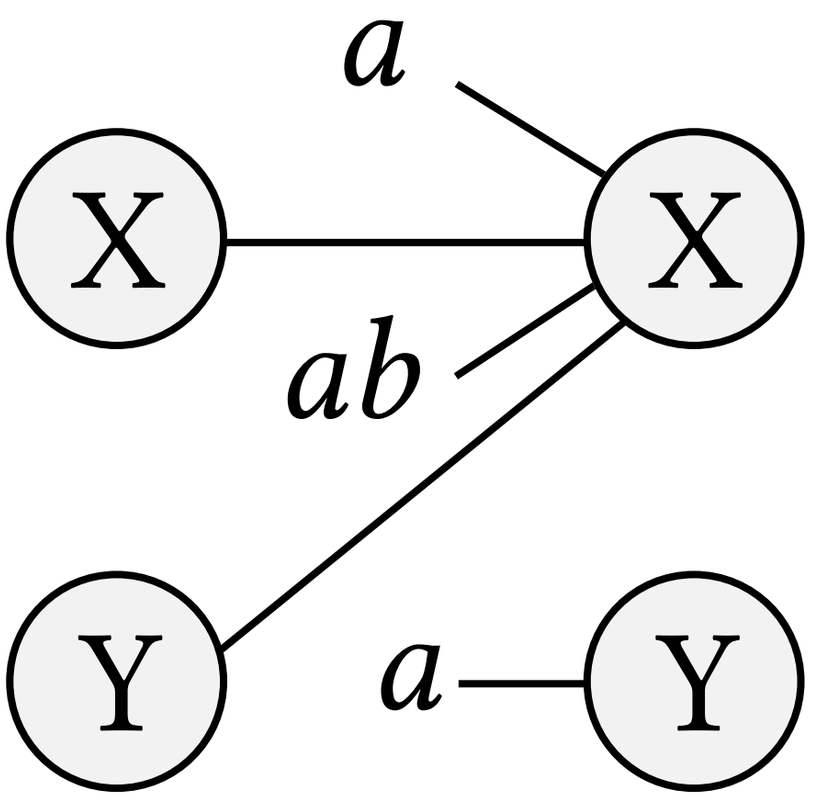

these columns are visited 3 times


x_{i-1}
x_i
x_{i+1}



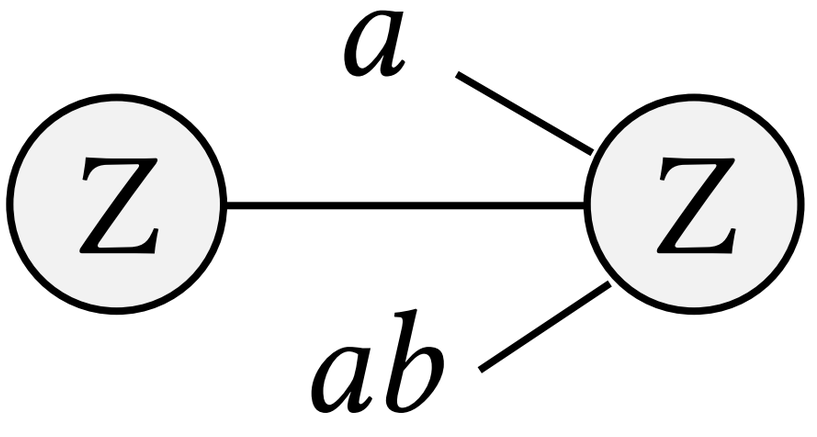

source/target
column
furthest
column


x_1
x_2
x_3
a
a
a
b
a
a
b
a
w_0
w_1
w_2
w_3

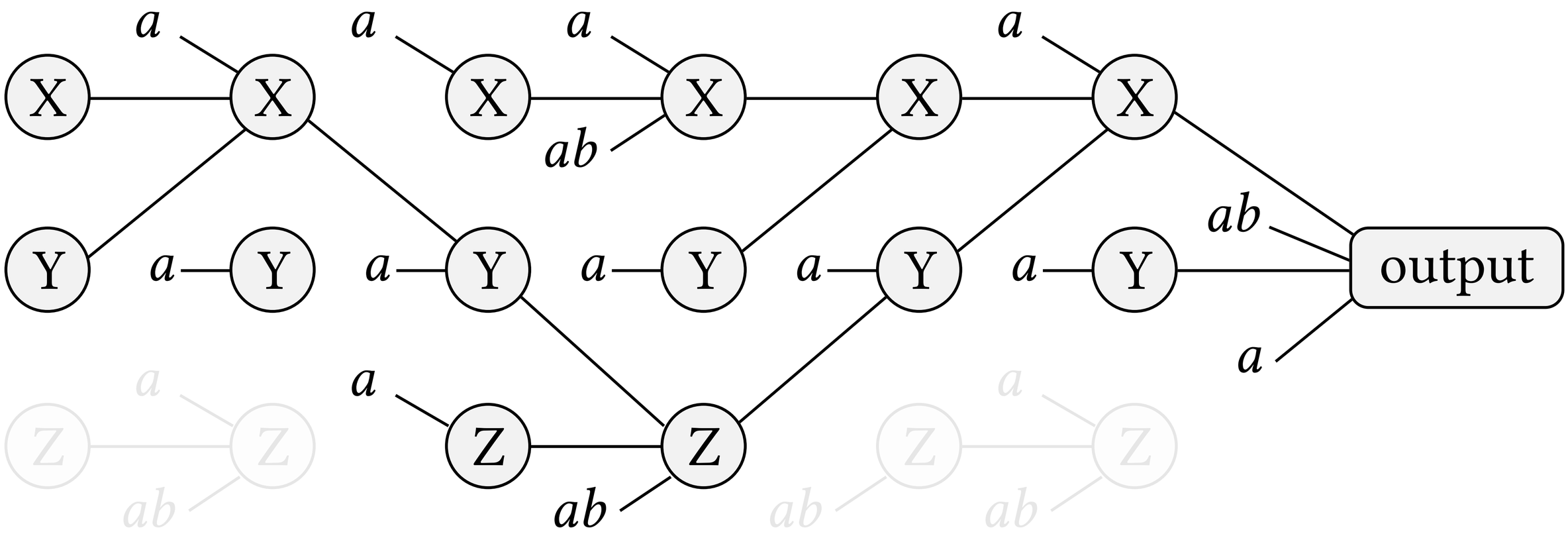


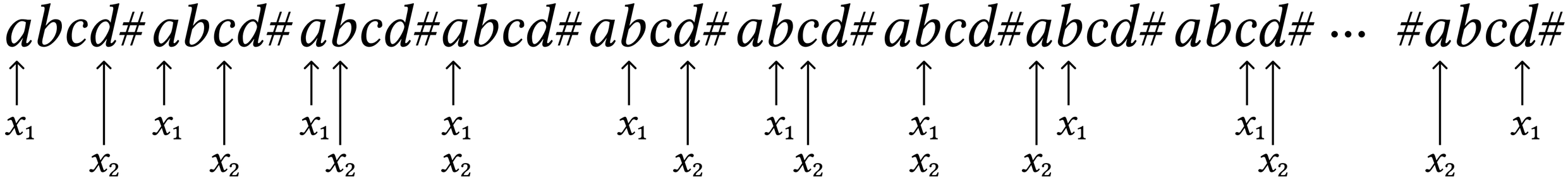
$\mathrm{X} := a\mathrm{X}ab\mathrm{Y}$

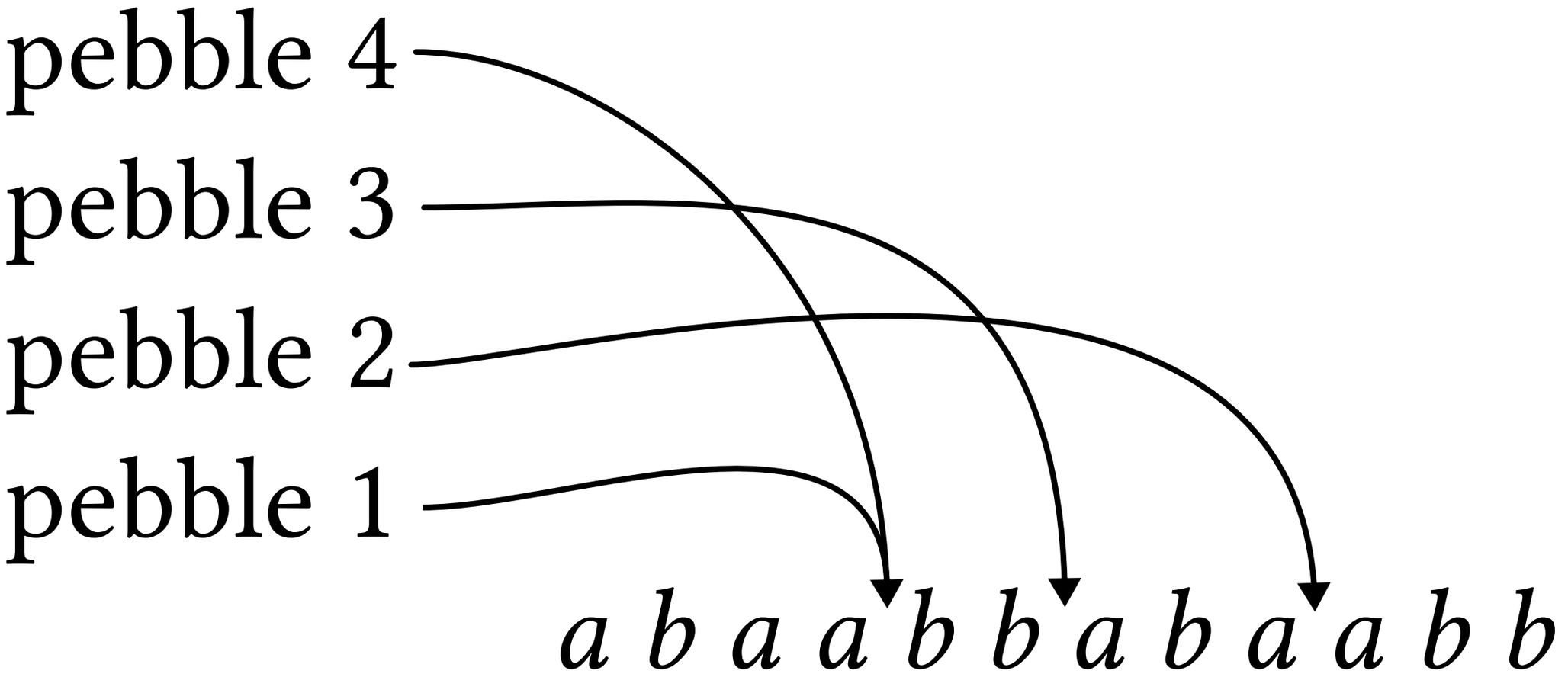
$\mathrm{Y} := a$

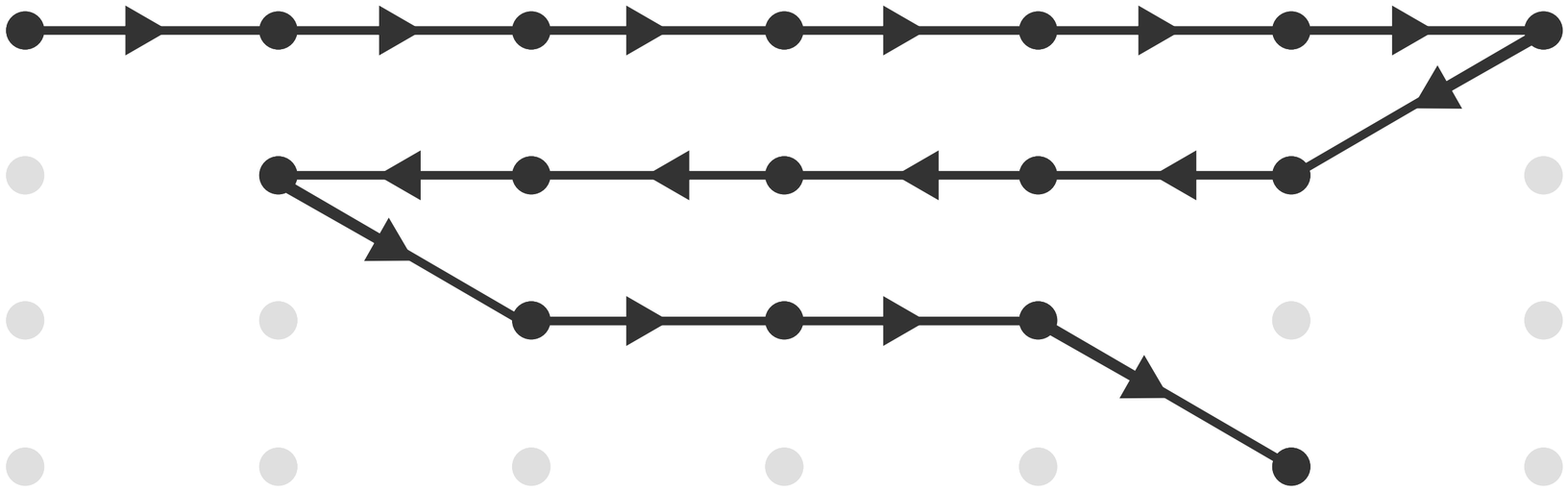


$\mathrm{Z} := a\mathrm{Z}ab$


a
a
a
a
X
X
X
X
X
ab
Y
a
Y
a
Y
a
Y
a
Y
ab
output
a
a
a
Z
Z
ab
ab
a
a
Z
Z
Z
Z
ab
ab


abcd# abcd# abcd# abcd# abcd# abcd# abcd# abcd# abcd# ... #abcd#
x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2 x1 x2


pebble 4
pebble 3
pebble 2
pebble 1
a b a a b b a b a a b b



movement of pebble $l+1$

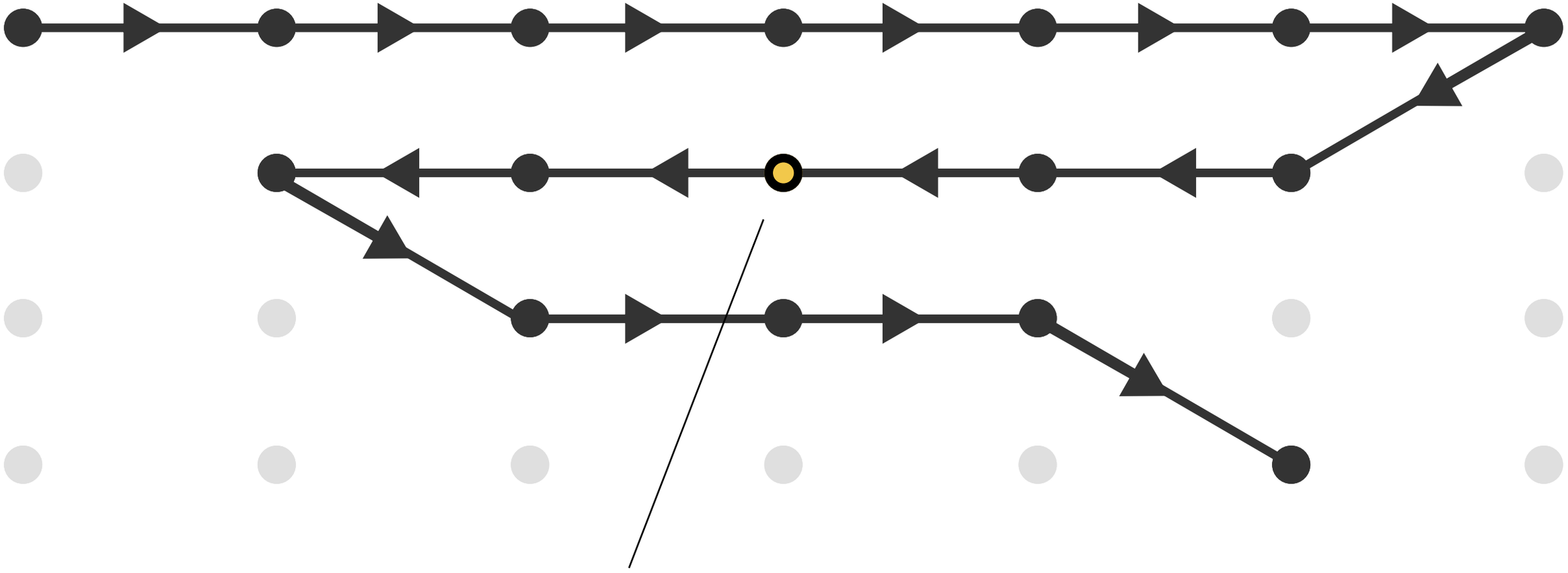
a_1 a_2 a_3 a_4 a_5 a_6

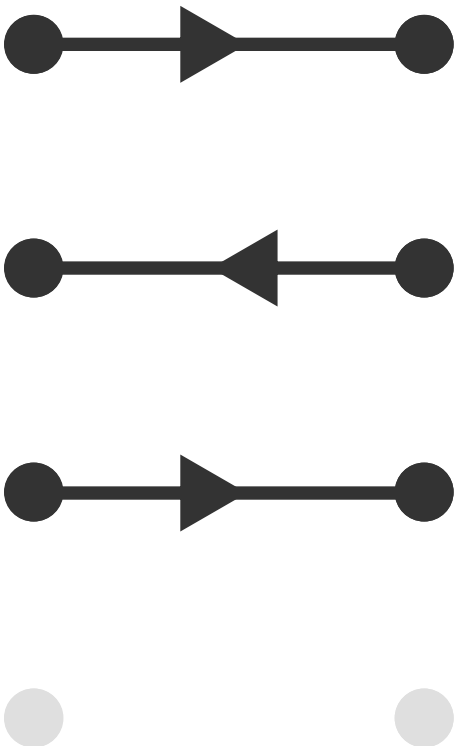
input string

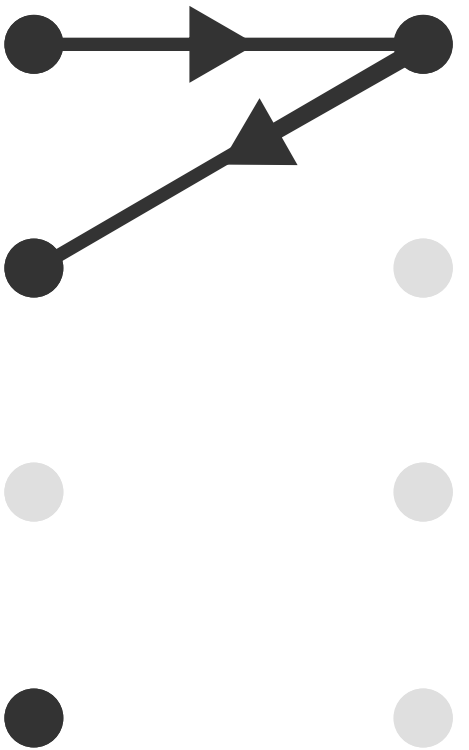
x_1 x_2

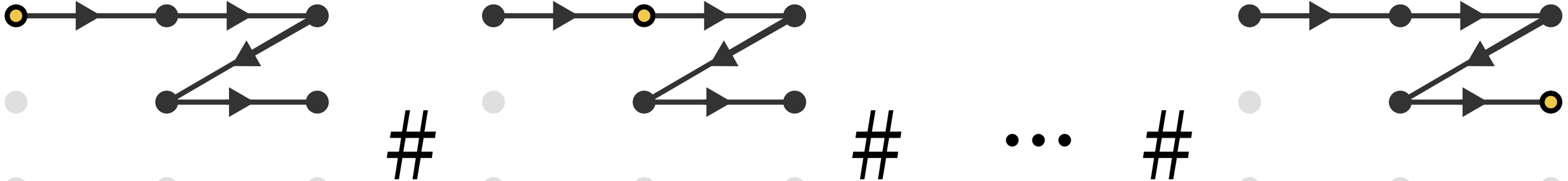
x_1
x_2

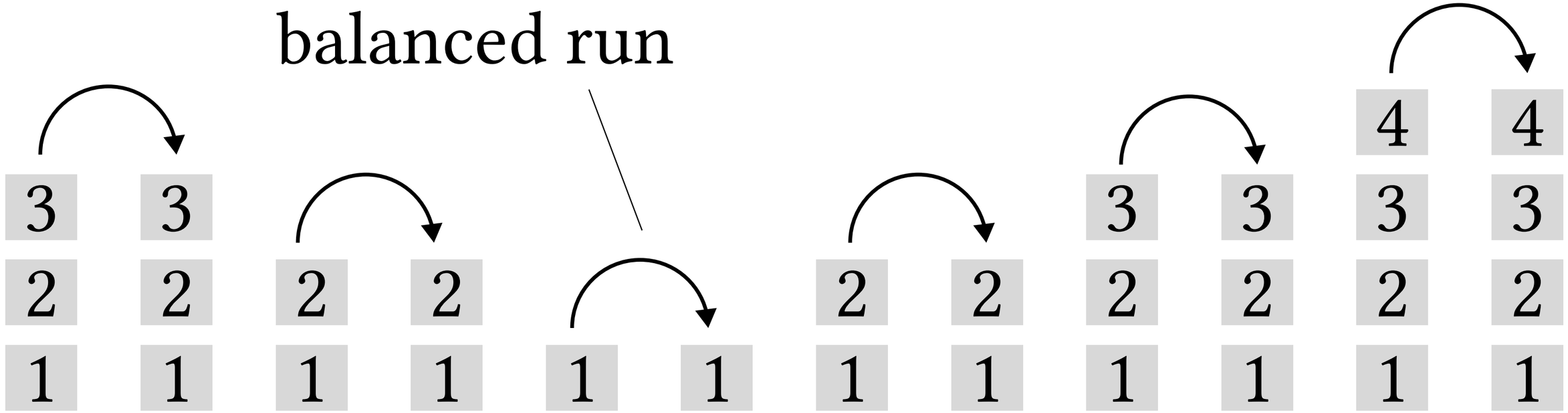
pebbles 1,...,l

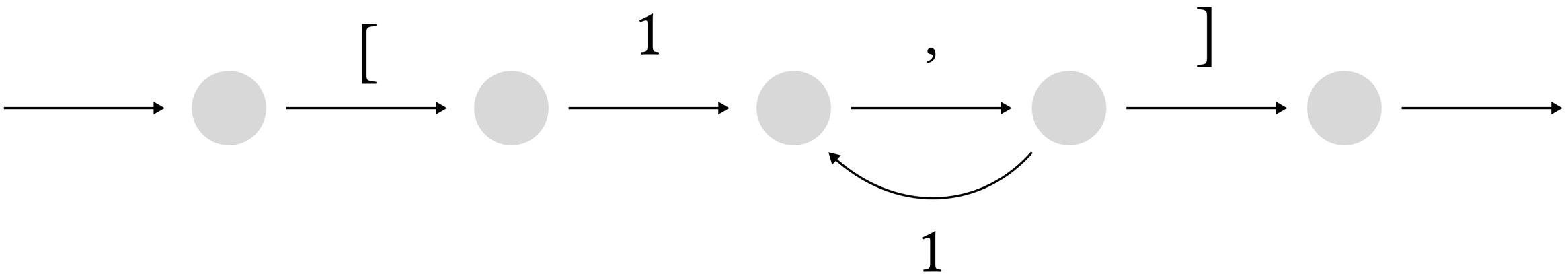

highlighted

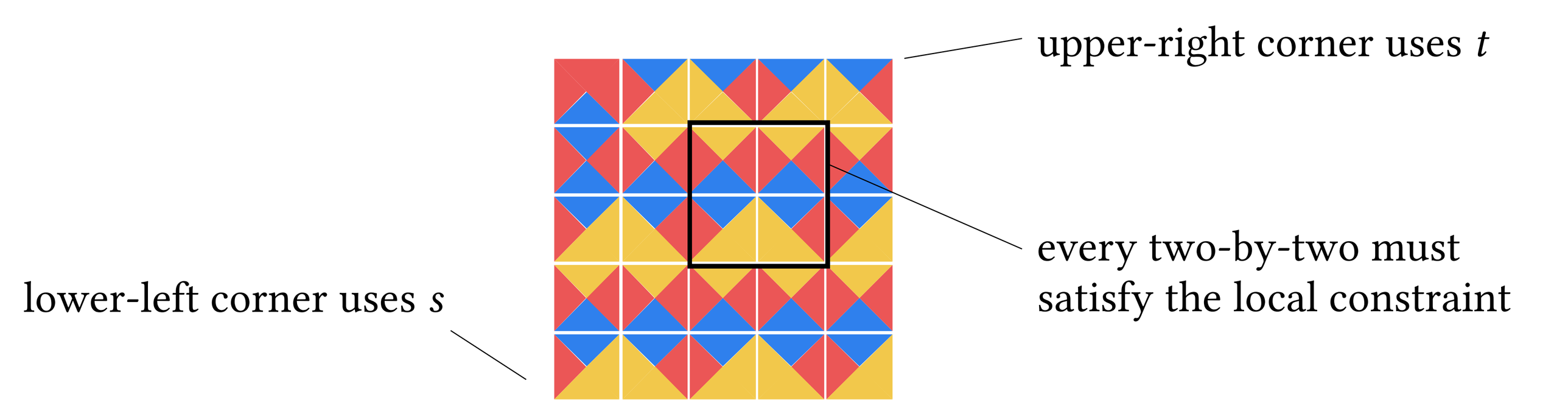


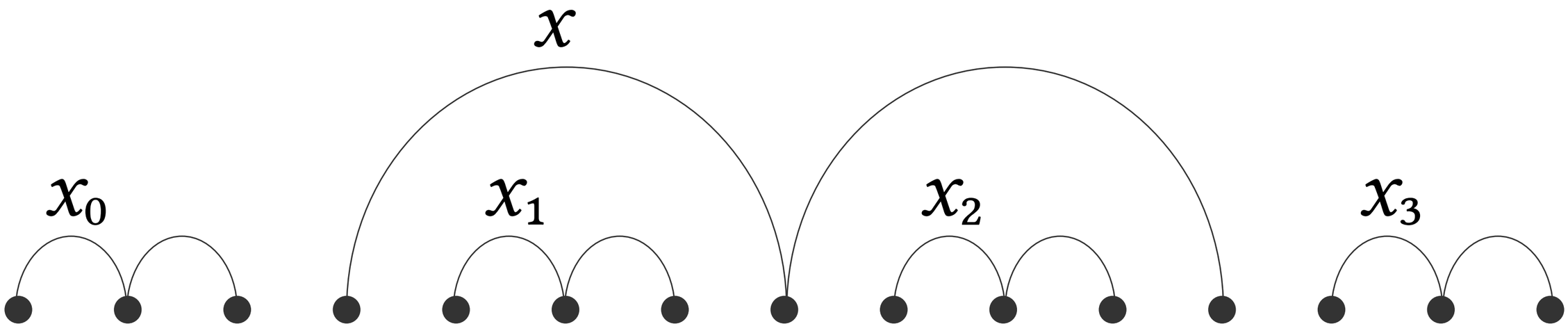


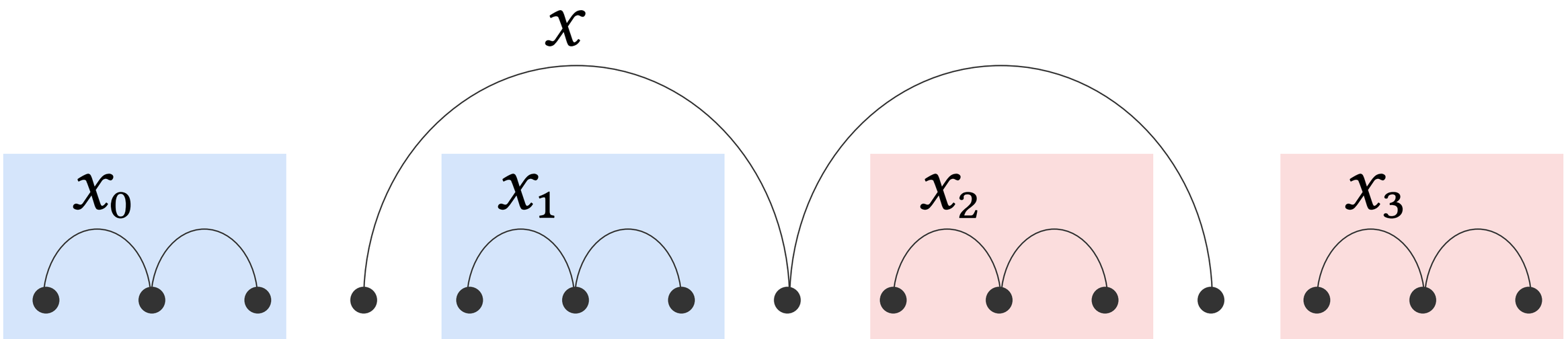


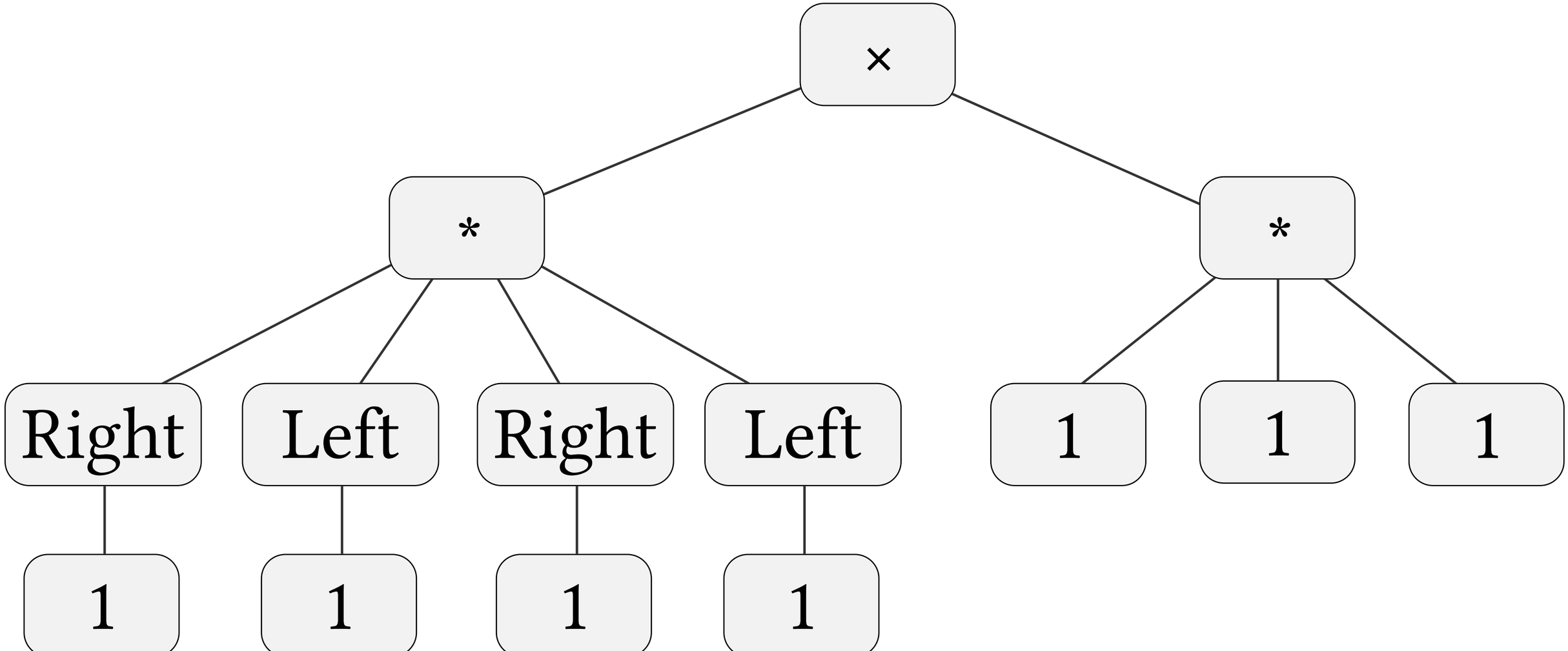

balanced run
3 3
2 2
1 1
2 2
1 1
1 1
2 2
1 1
3 3
2 2
1 1
4 4
3 3
2 2
1 1


[
1
,
]
1

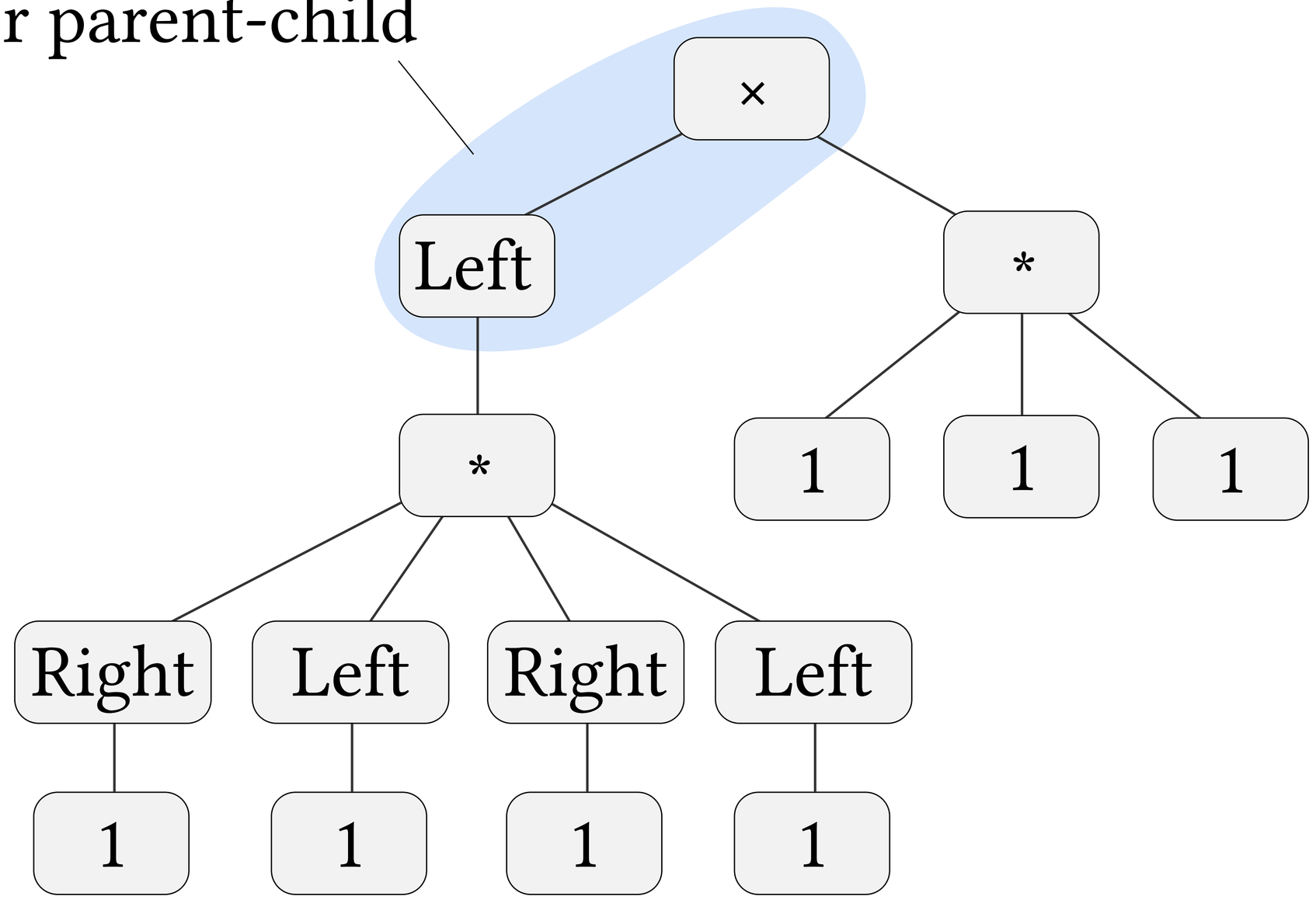

upper-right corner uses t
every two-by-two must
satisfy the local constraint
lower-left corner uses s


x
x_0
x_1
x_2
x_3


x
x_0
x_1
x_2
x_3
smaller than x
bigger than x


×
*
*
Right
Left
Right
Left
1
1
1
1
1
1
1

a binary relation
for parent-child



if we want 3log$n$ bits,
then we have 3 pebbles



each pebble represents a
part of the stack

one stack for the left
part of the memory tape

one stack for the right
part of the memory tape

monoid elements assigned by the homomorphism to the maximal factors without distinguished positions

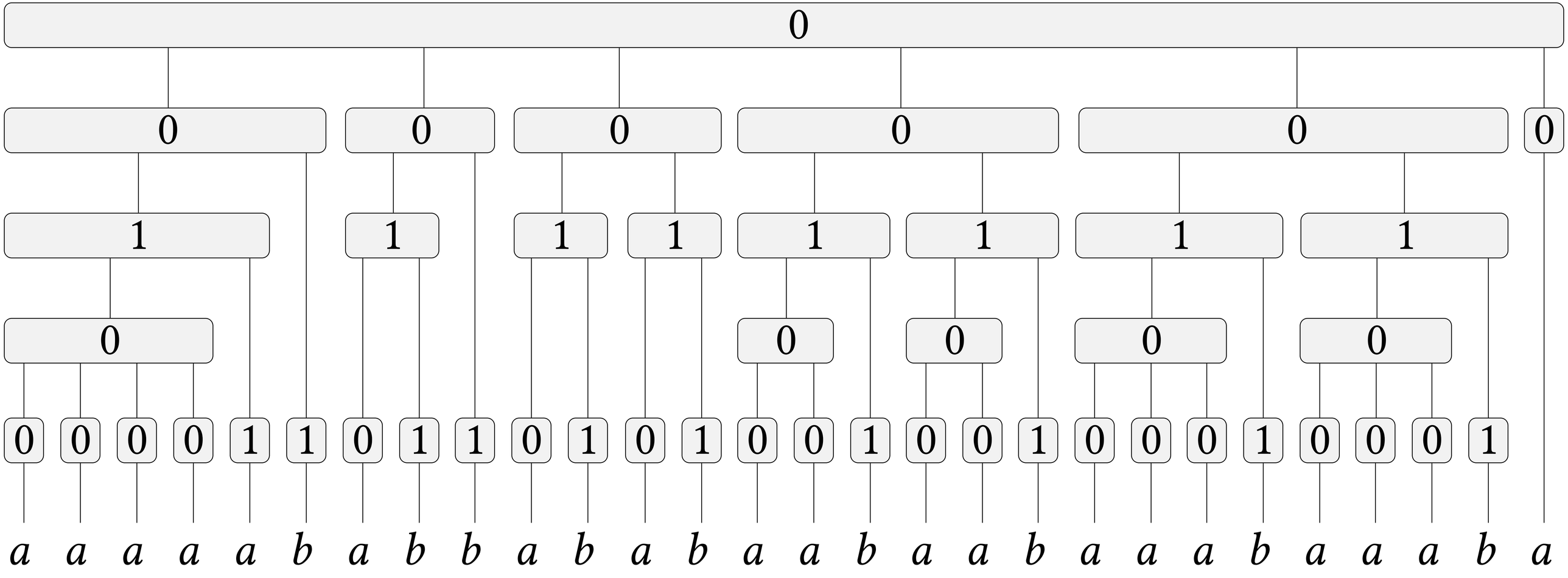
$aaabaabbaab\ b\ abbbaaabaaba\ aabbaaabaabb\ b\ aabbabbb$

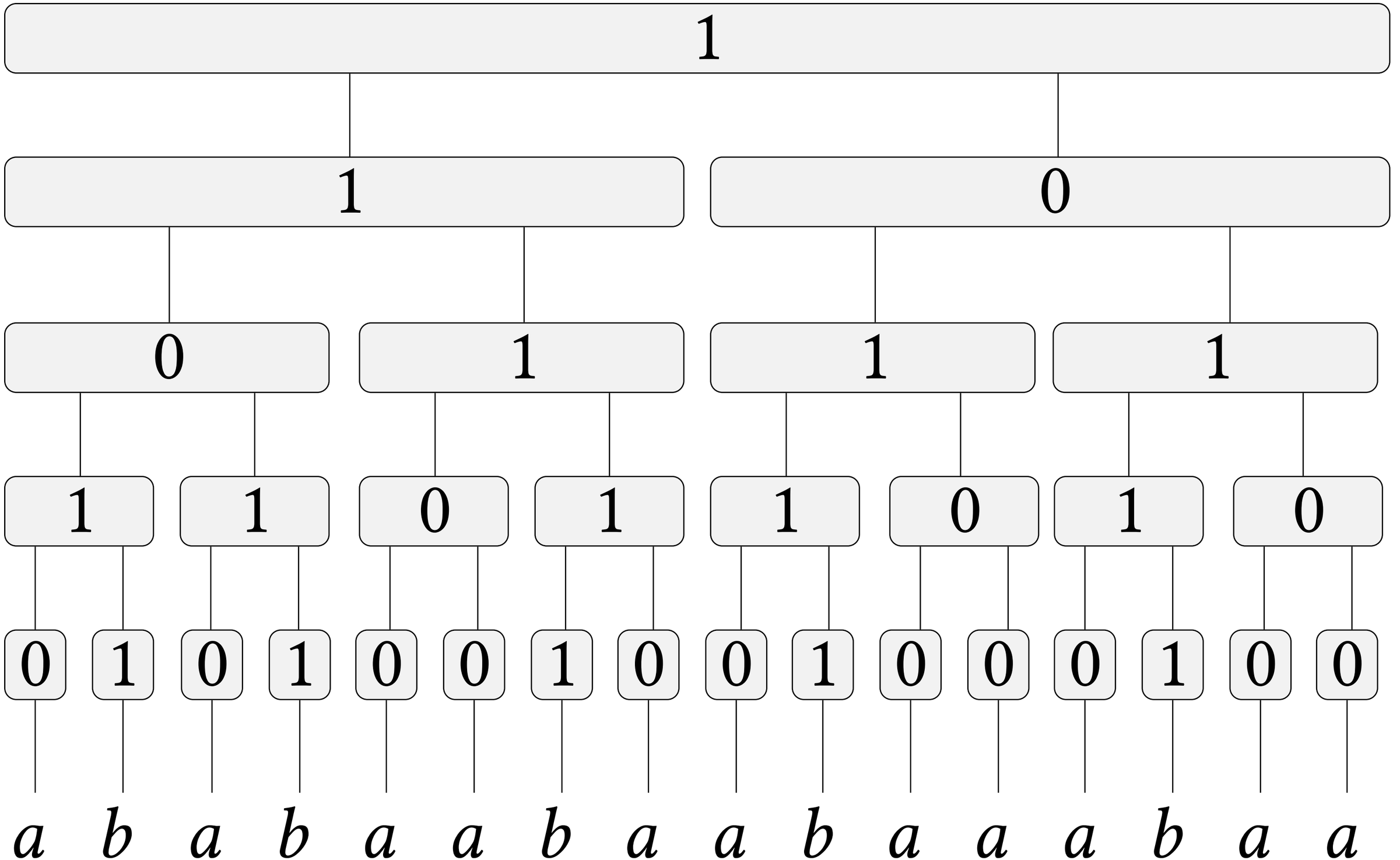
x_1
x_3

f

x_2

the order on distinguished positions, and their labels


0
0 0 0 0 0 0
1 1 1 1 1 1
0 0 0 0 0
0 0 0 0 1 1 0 1 1 0 1 0 1 0 0 1 0 0 1 0 0 0 1 0 0 0 1
a a a a a b a b b a b a b a a b a a b a a a b a a a b a


1
1
0
0
1
1
1
0
1
1
1
0
1
1
0
1
1
0
1
0
0
1
0
0
1
0
0
1
0
0
0
1
0
0
a
b
a
b
a
a
b
a
a
b
a
a
a
b
a
a

1

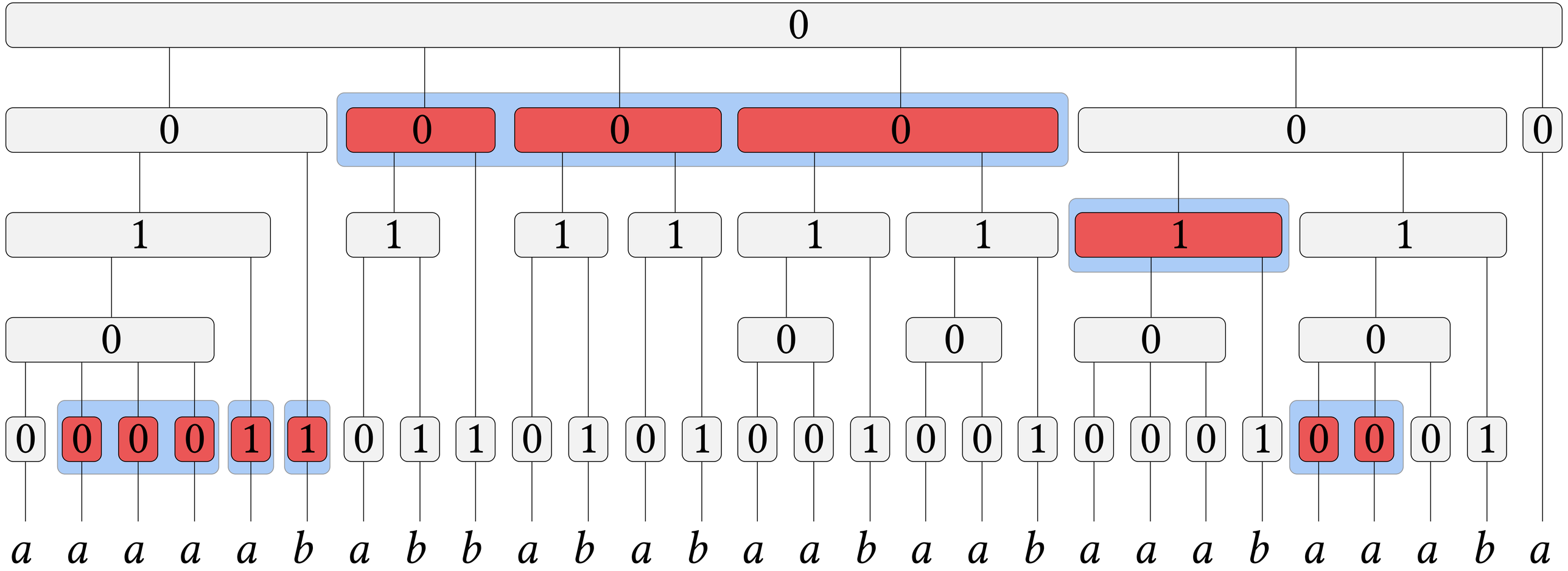
1 0

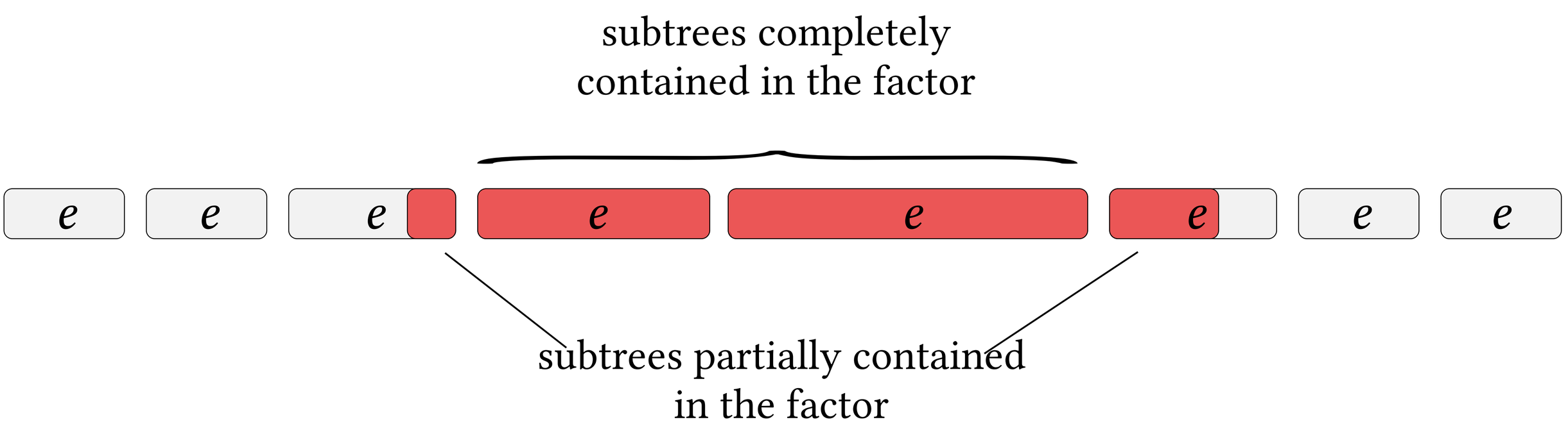
0 1 1 1

1 1 0 1 1 0 1 0

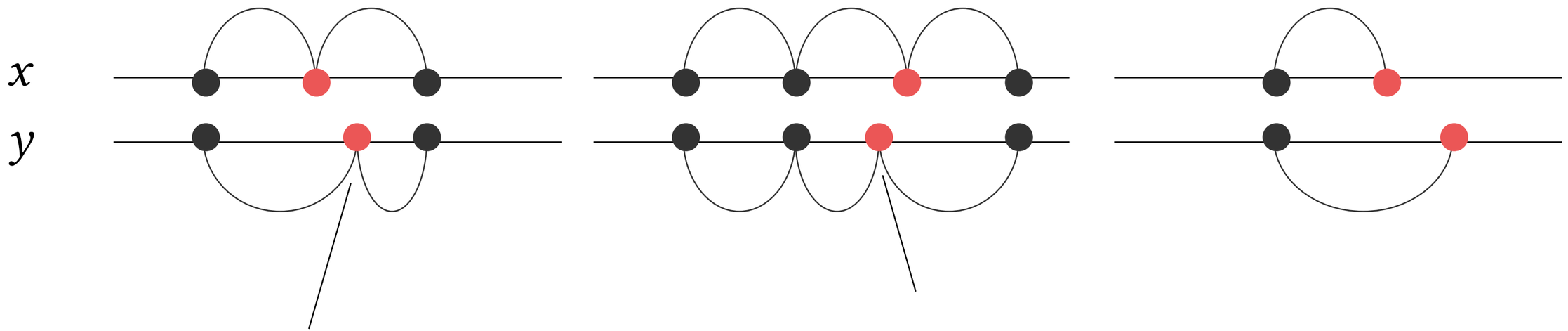
0 1 0 1 0 0 1 0 0 1 0 0 0 1 0 0

a b a b a a b a a b a a a b a a


0
0 0 0 0 0
1 1 1 1 1 1 1
0 0 0 0 0
0 0 0 0 1 1 0 1 1 0 1 0 1 0 0 1 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1
a a a a a b a b b a b a b a a b a a b a a a b a a a b a

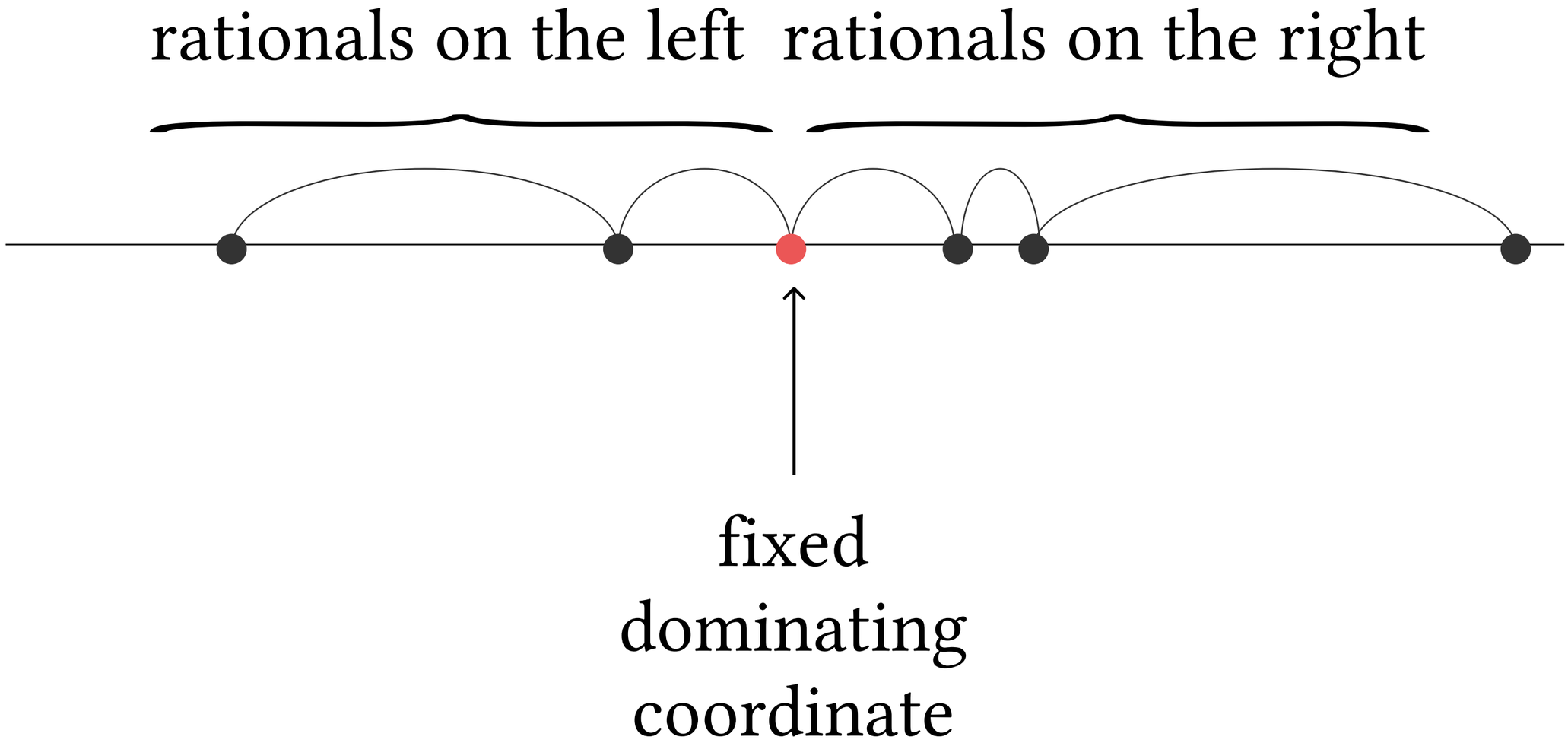

subtrees completely
contained in the factor
e
e
e
e
e
e
e
e
subtrees partially contained
in the factor

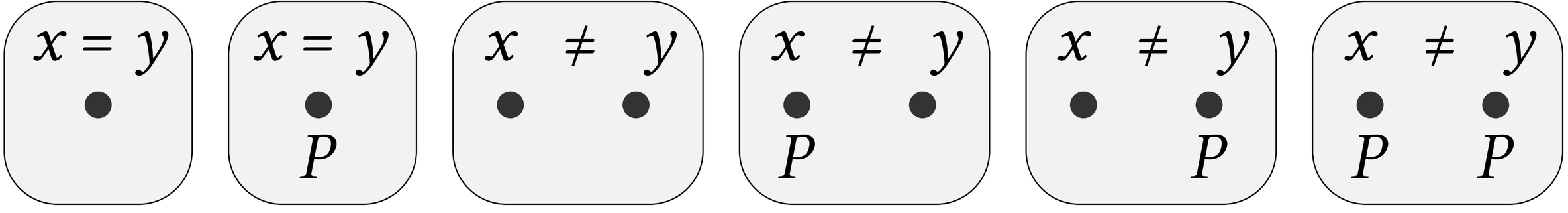
Q_1 Q_2 Q_3

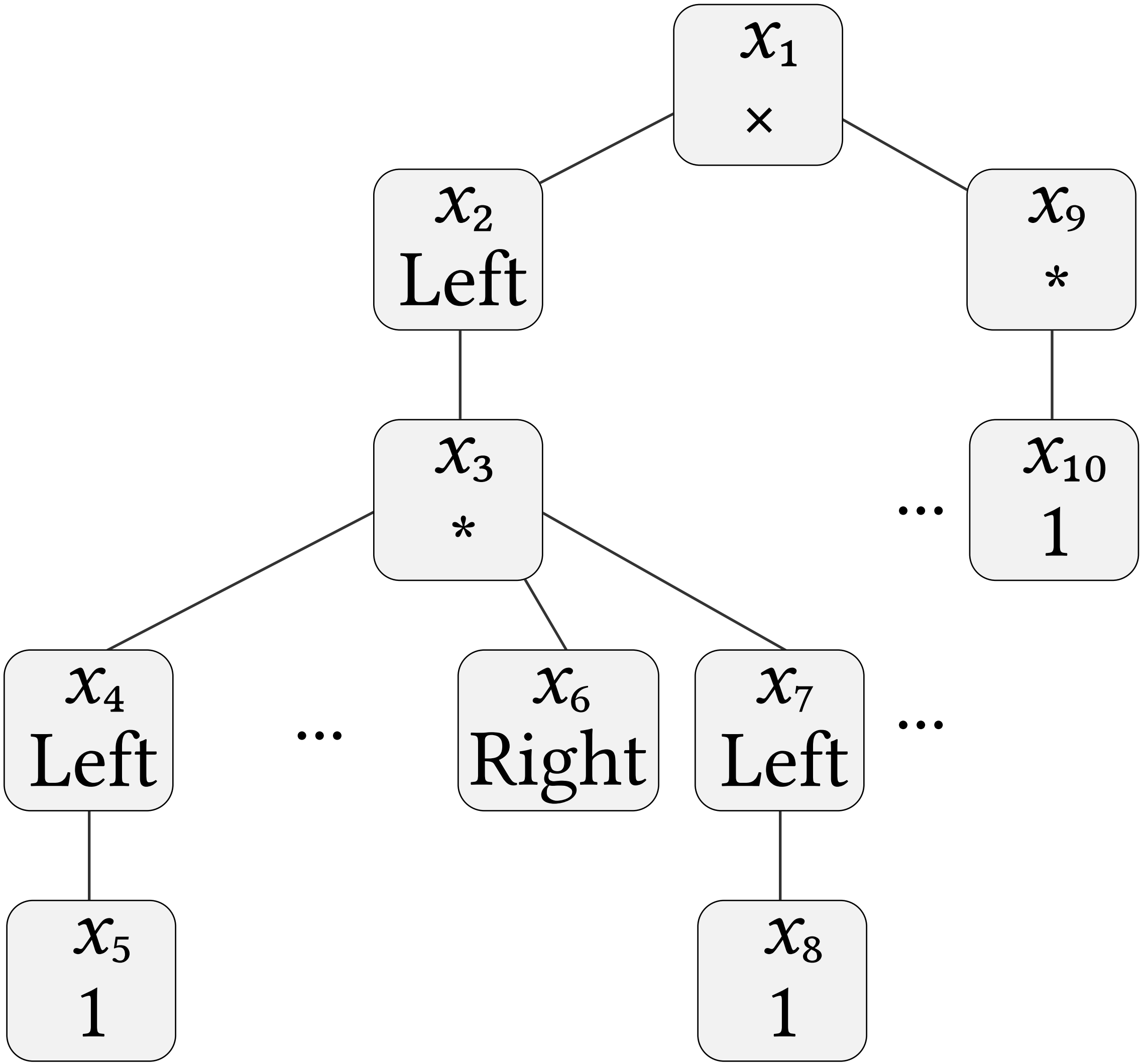


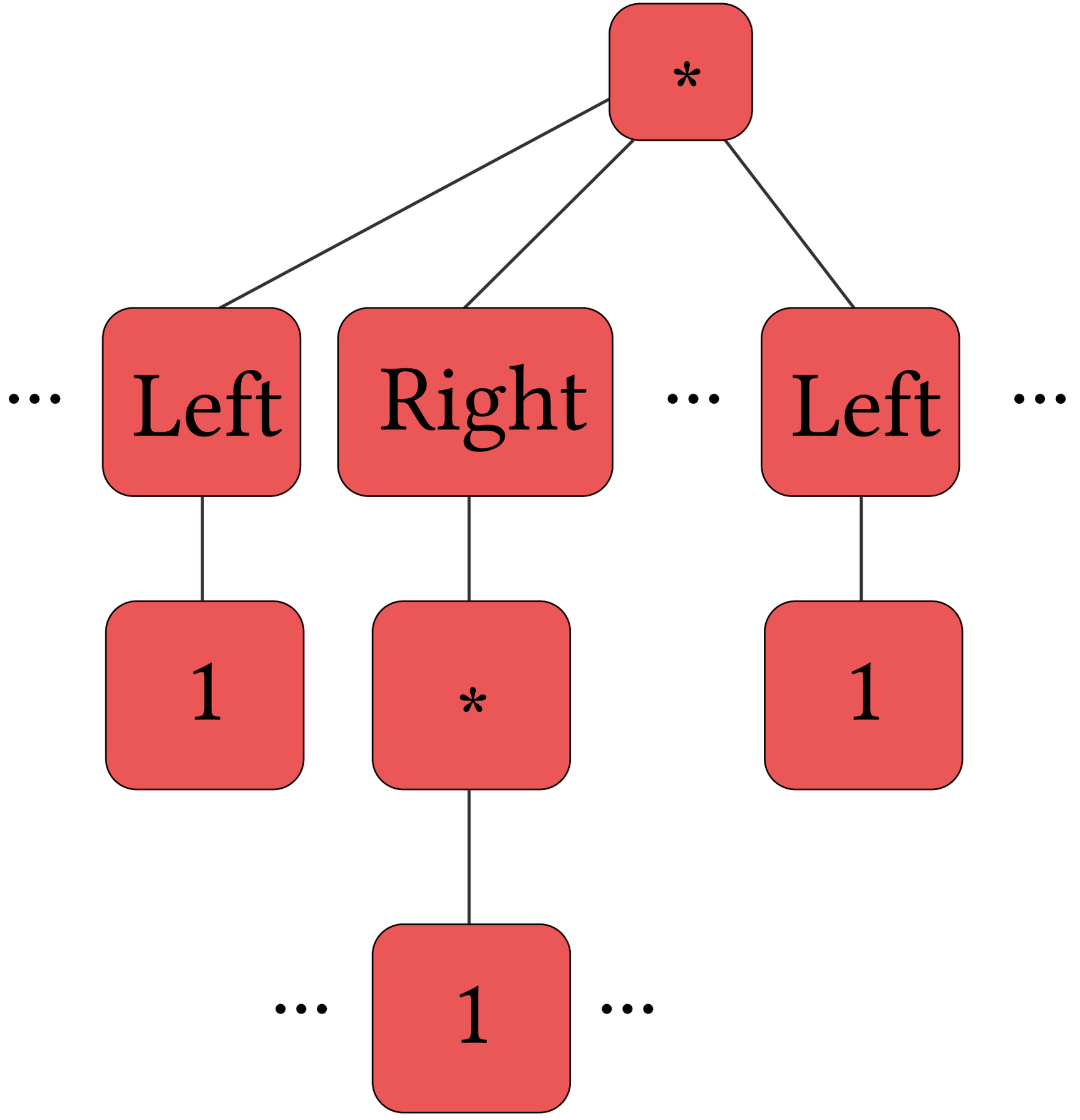
if $i \in I$, then we move the dominating coordinate to the right, and keep the non-dominating coordinates unchanged.

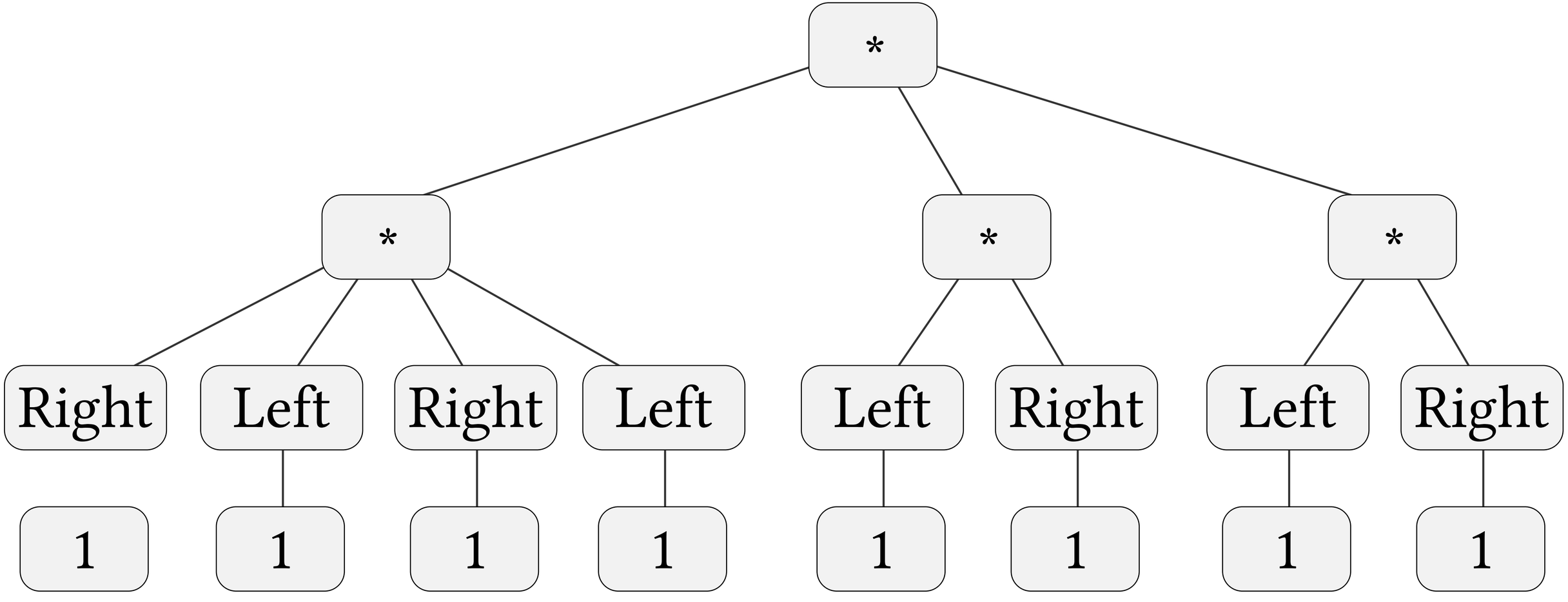
if $i \notin I$, then we move the dominating coordinate to the left, and keep the non-dominating coordinates unchanged.

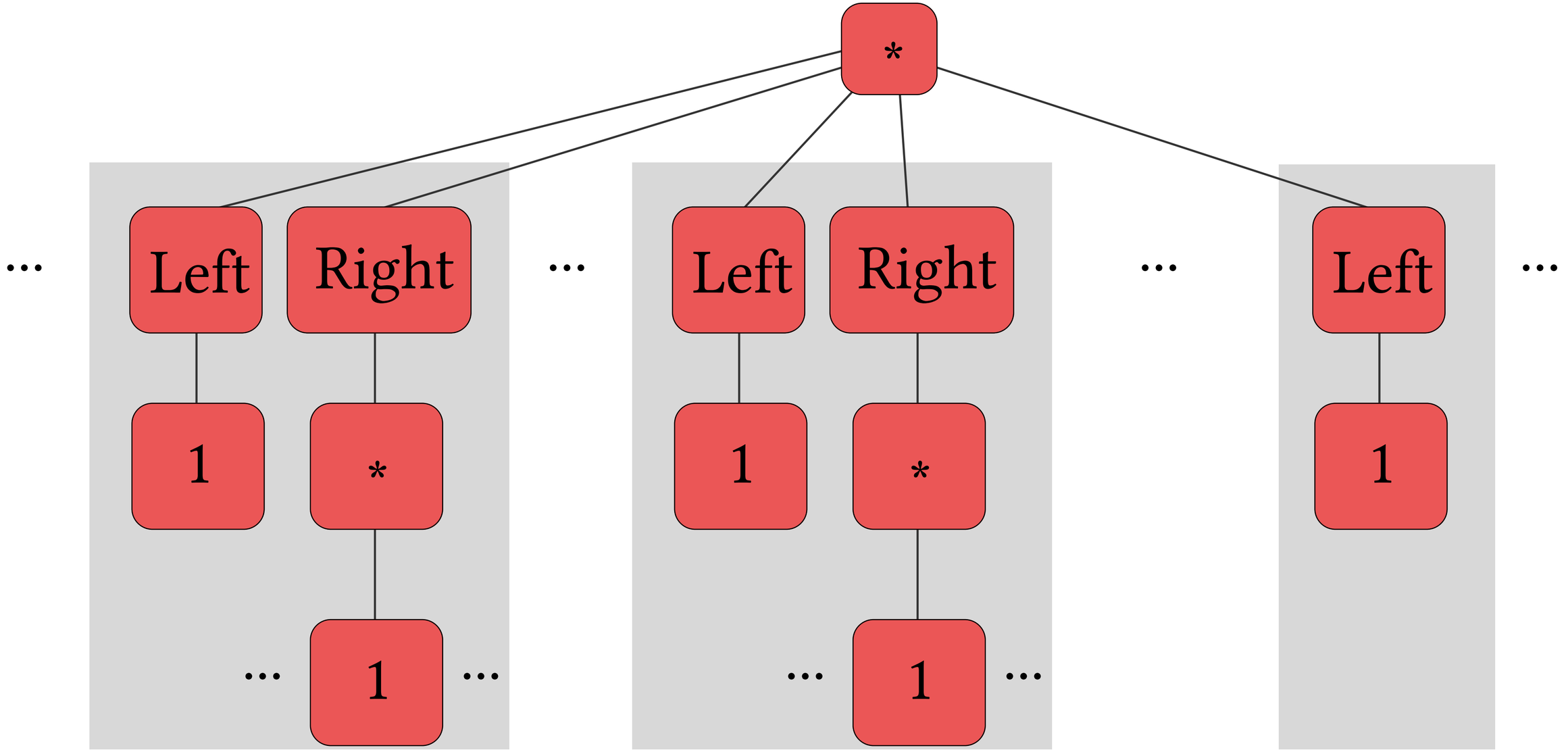

rationals on the left
rationals on the right
fixed
dominating
coordinate


x = y
x = y
P
x ≠ y
x ≠ y
P
x ≠ y
P
x ≠ y
P P


x_1
×
x_2
Left
x_9
*
x_3
*
x_{10}
1
x_4
Left
x_6
Right
x_7
Left
x_5
1
x_8
1


*
...
Left
Right
...
Left
...
1
*
1
...
1
...

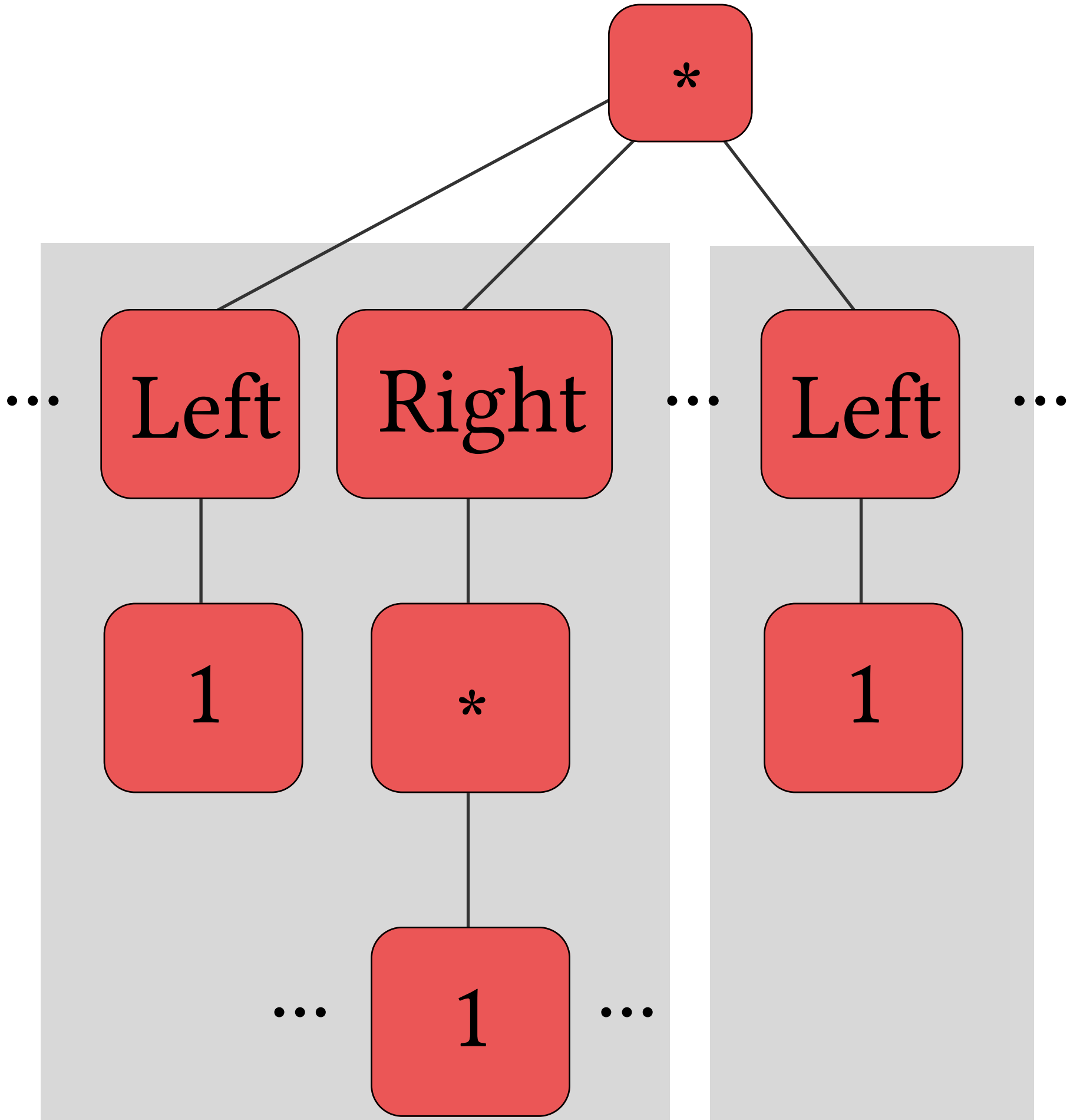

*
*
*
*
Right
Left
Right
Left
Left
Right
Left
Right
1
1
1
1
1
1
1
1

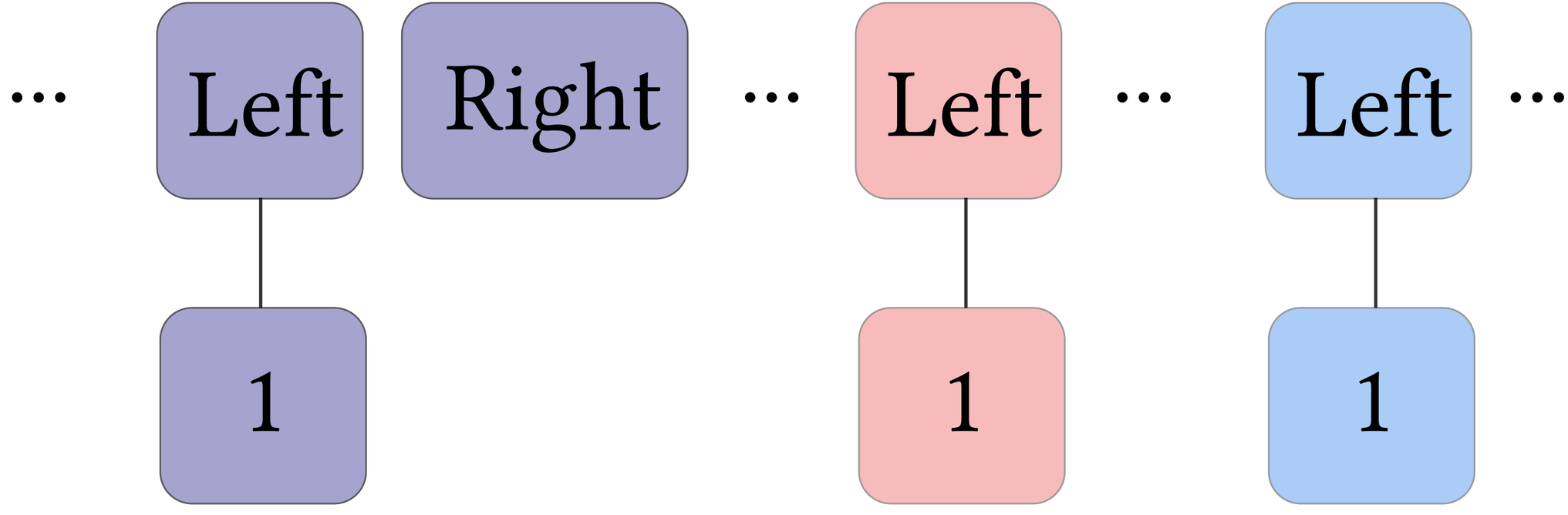


segment 1 in x

segment 1 in y

segment 2 in x
=
segment 3 in y


*
Left
Right
Left
1
*
1
1
segment 1
segment 2



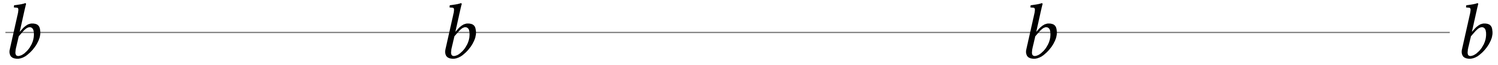
violet nodes are
from both x and y



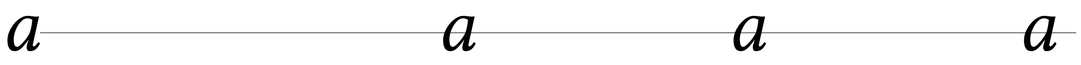
output string:



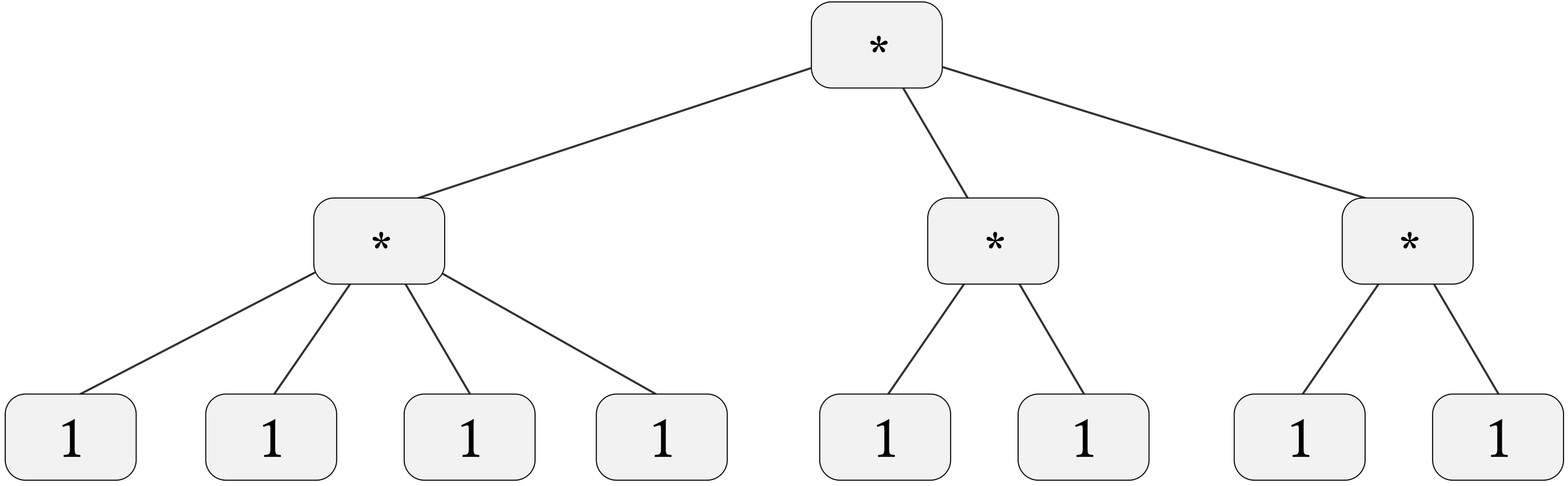
output from arrangement type 1

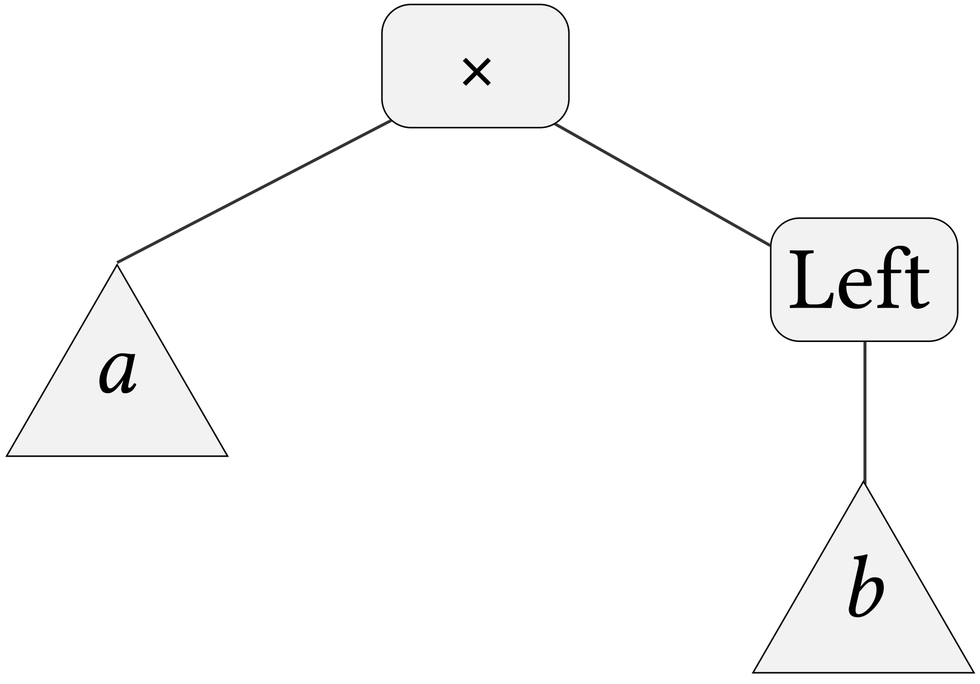


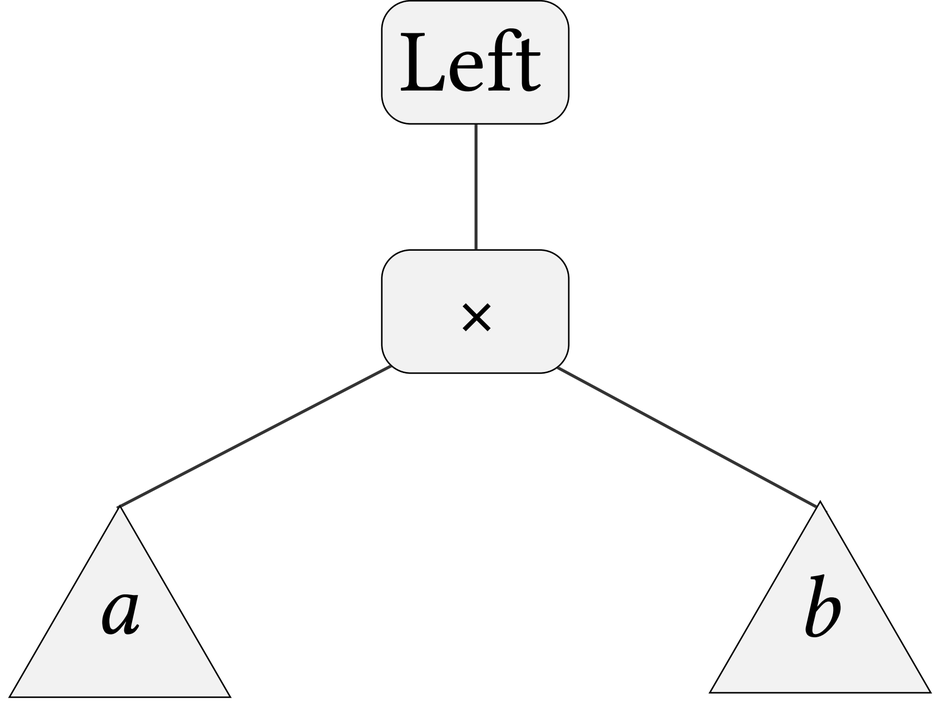
output from arrangement type 2

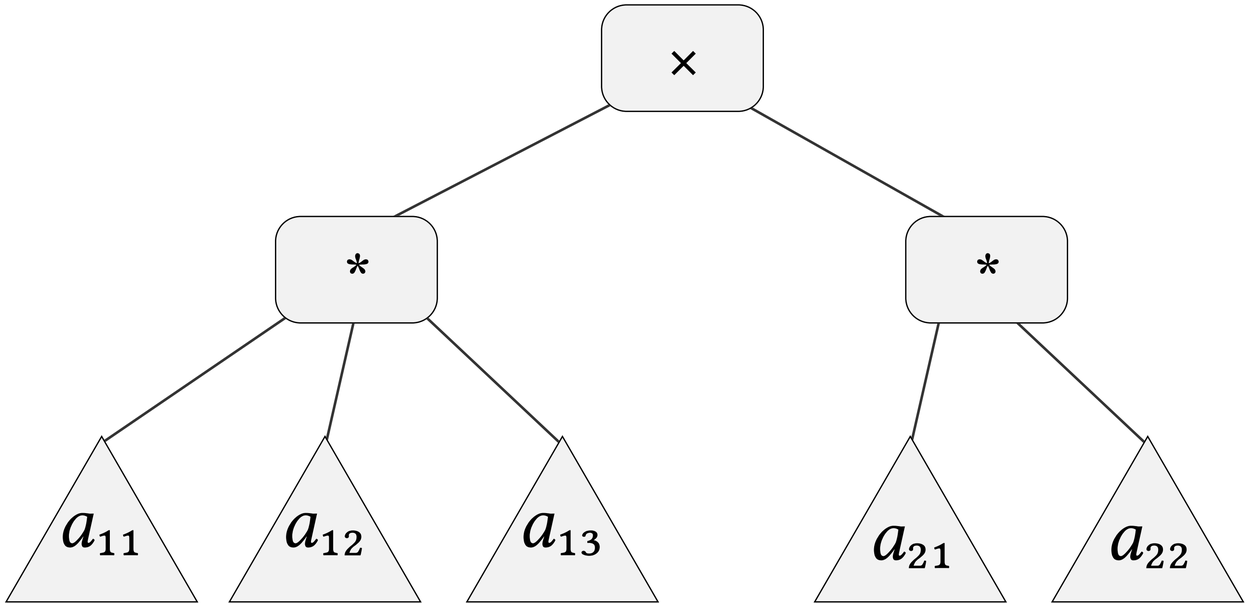


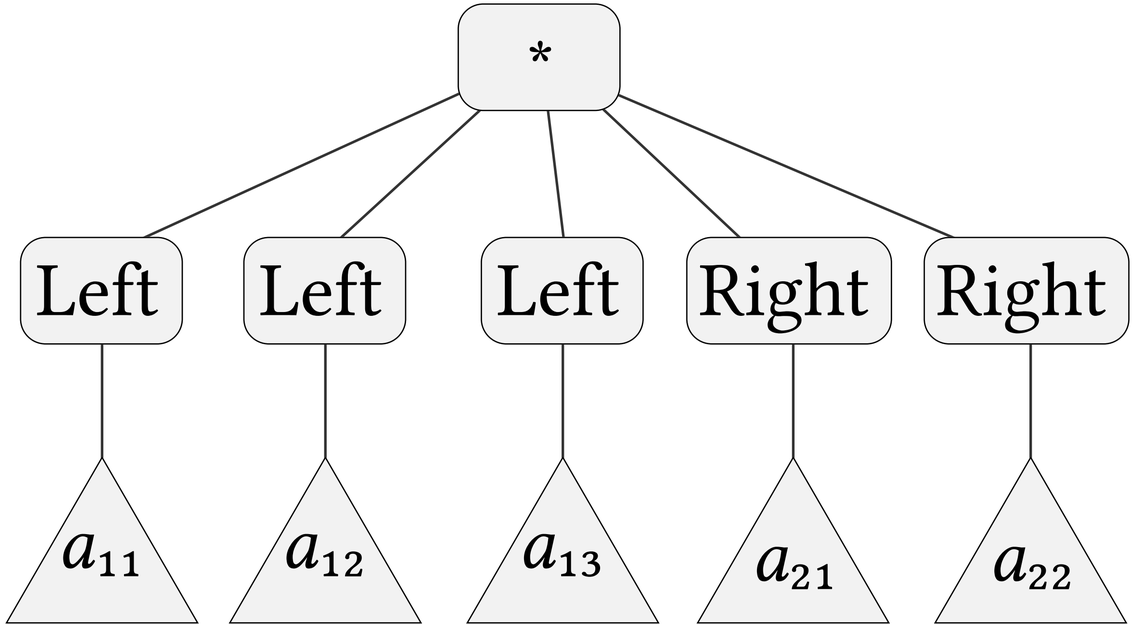
output from arrangement type 3

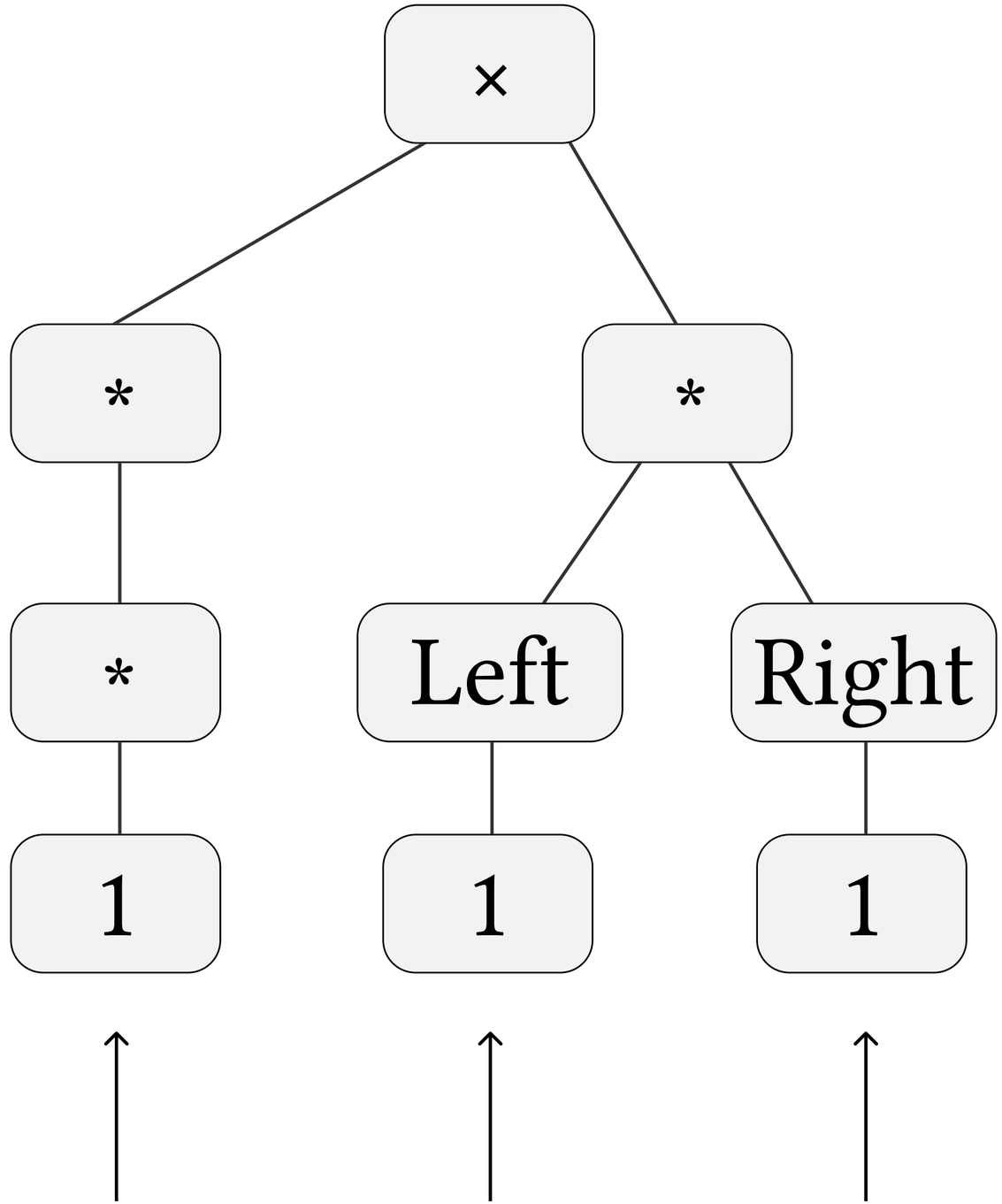

*
*
*
*
1
1
1
1
1
1
1
1


×
Left
a
b

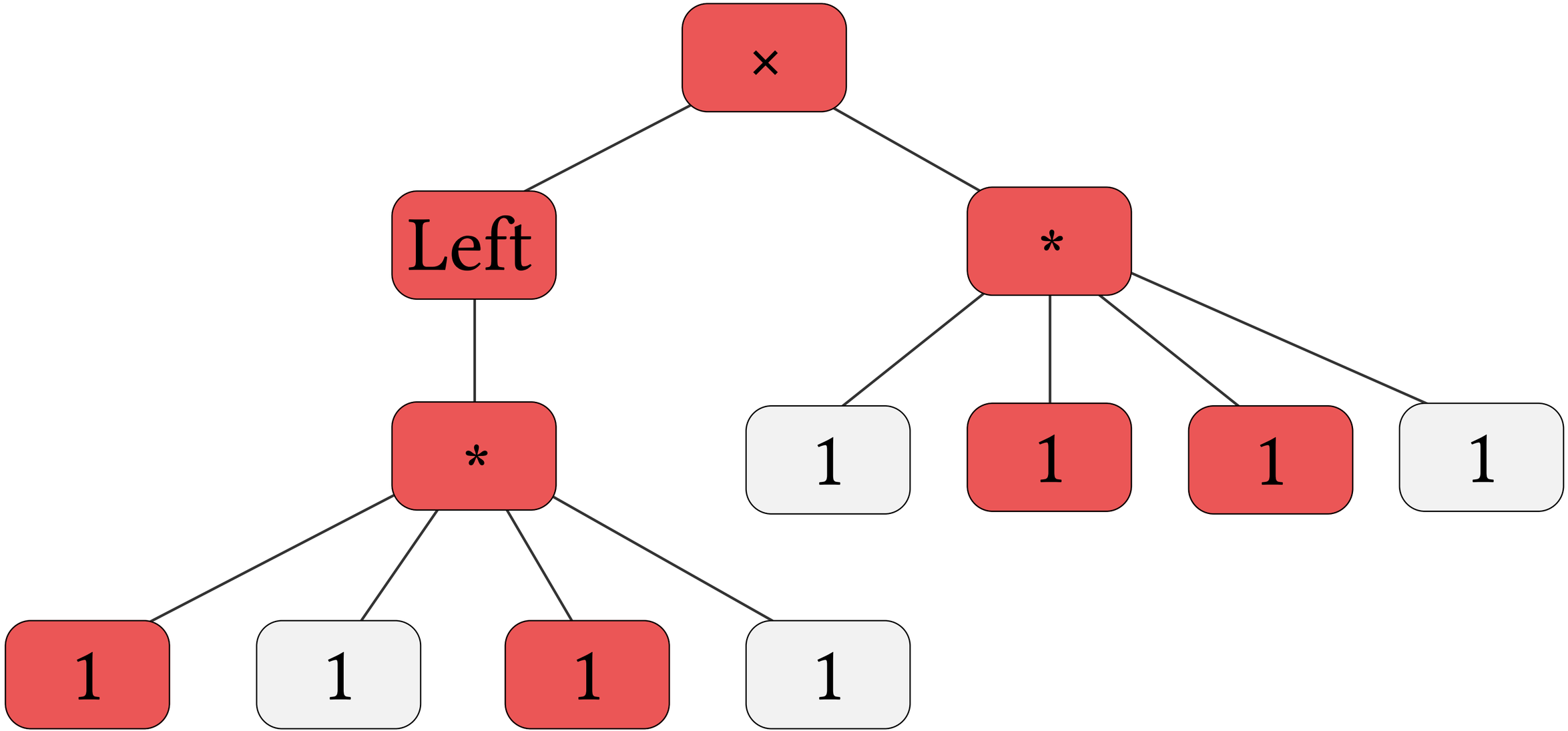

Left
×
a
b

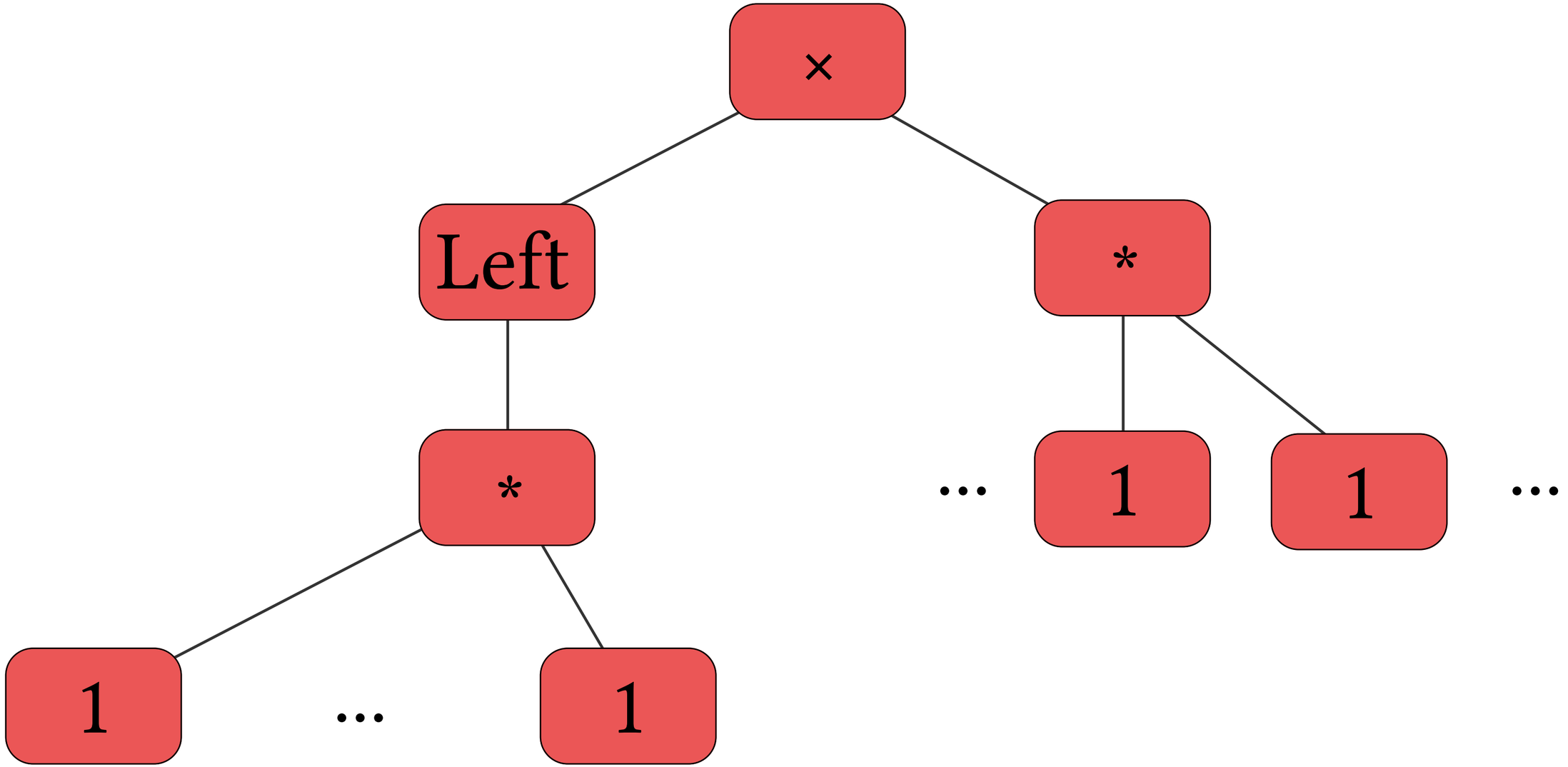

×
*
*
a_{11}
a_{12}
a_{13}
a_{21}
a_{22}


*
Left
Left
Left
Right
Right
a_{11}
a_{12}
a_{13}
a_{21}
a_{22}



parameter is 4 = 2 + 1 + 1


×
Left
*
*
1
1
1
1
1
1
1
1


×
Left
*
*
1
…
1
1
…
1
…